Logic

Aria Ruler(Forouzandeh)

March 30, 2026

This page intentionally left blank.

Contents

Chapter 1

Atomes

1.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 1.1.1. ***Urelements*** *are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.*

In this book, we will denote individual urelements using lowercase letters u, v, w,...

Undefined 1.1.2. ***Type*** *parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).*

1.	u		
2.		v	
3.			w

Rule 1.1.1. ***fusion*** *it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)*

$$\begin{array}{r|l|l}
m. & u_0 & \\
 & \vdots & \\
m+n. & u_n & \\
o. & & \{u_0, u_1 ... u_n\}
\end{array}$$

Undefined 1.1.3. ***Set*** *we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).*

definition 1.1.1. ***Object*** *urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).*

Rule 1.1.2. ***fission*** *it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)*

$$\begin{array}{r|l|l}
o. & & \{a_0, a_1 ... a_n\} \\
m. & a_0 & \\
 & \vdots & \\
m+n. & a_n &
\end{array}$$

Example 1.1.1. $\{a, b, c\} / \therefore \{a, b\}$ *(prove if $\{a, b, c\}$ exists then $\{a, c\}$ exists)*

1.		$\{a, b, c\}$	
2.	a		*fi1*
3.	b		*fi1*
4.	c		*fi1*
5.		$\{a, c\}$	*fu2,3,4*

Example 1.1.2. $\{a, b\}/\therefore \{b, a\}$

1.		$\{a, b\}$	
2.	a		*fi1*
3.	b		*fi1*
4.		$\{b, a\}$	*fu2,3*

This page intentionally left blank.

Chapter 2

propositional logic (zeroth-order logic)

2.1 What is proposition?

definition 2.1.1. ***predicate*** *"set()" and identity "=" and membership "∈" symbols are predicates. "set()" is unary and identity and membership are binary.*

capital letter "(P, Q, R)" showes **predicates**.

definition 2.1.2. ***Atomic proposition*** *we define atomic propositions like this.*

1. *if "a" is an obj then "set(a)" is an atomic proposition.*
2. *if "a, b" are obj then "$a = b$" is an atomic proposition.*
3. *if "a, b" are obj then "$a \in b$" is an atomic proposition.*

we denote atomic propositions with "p_i".

Undefined 2.1.1. ***logical operators*** *$(\neg, \to, \wedge, \vee)$ tool that used between atoms to make more complex propositions (all of the opratores are binary except "¬" that is unary).*

you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.

definition 2.1.3. ***Propositions*** *these are propositions.*

1. *atomic propositions "(p_i)".*
2. *if "p" is a formula then "$\neg p$" is a formula too.*
3. *if "p, q" are two propositions then "$p \bigcirc q$" is a formula too.($\bigcirc$ can be "$\to, \wedge, \vee$").*

for example "$(p_0 \to p_1, \neg p_3)$"
we denote propositions with "$p, q, r, \ldots$".

2.2 Rules of propositions.

definition 2.2.1. ***proposition schema(variable proposition)*** *a proposition that is unknown for us and it works as variable.(we show them with "$\varphi, \psi, \theta, ...$")*

Undefined 2.2.1. ***Assumption for conditional proof(acp)*** *is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shoter than left one.(it can't fall out). and every hypothesis that opens must be closed.)*

$$
\begin{array}{l|l|l|l}
1 & & & \\
2 & & \multicolumn{1}{|l}{} & \\
3 & & \multicolumn{1}{|l}{} & \multicolumn{1}{|l}{} \\
4 & & \multicolumn{1}{|l}{} & \multicolumn{1}{|l}{} \\
5 & & \multicolumn{1}{|l}{} & \\
6 & & &
\end{array}
$$

Rule 2.2.1. ***rules of logical operations*** ($\to I, \to E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \bot I, \bot E$) *this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)*

$\begin{array}{\|l} \varphi \\ \vdots \\ \psi \end{array}$ $\therefore \varphi \to \psi \qquad \to I$	$\varphi \to \psi$ $\underline{\varphi}$ $\therefore \psi \qquad \to E$

Example 2.2.1. *for "p, q, r" we can prove these.*

1. $p \to q, q \to r / \therefore p \to r$
2. $q / \therefore p \to q$

$$
\begin{array}{lll}
1 & p \to q & ass \\
2 & q \to r & ass \\
3 & \quad\mid p & acp \\
4 & \quad\mid q & \to E1,3 \\
5 & \quad\mid r & \to E2,4 \\
6 & p \to r & \to I
\end{array}
\qquad\qquad
\begin{array}{lll}
1 & q & ass \\
2 & \quad\mid p & acp \\
3 & \quad\mid q & ass \\
4 & p \to q & \to I
\end{array}
$$

$\begin{array}{l} \varphi \\ \underline{\psi} \\ \therefore \varphi \wedge \psi \qquad \wedge I \end{array}$	$\begin{array}{l} \underline{\varphi \wedge \psi} \\ \therefore \varphi \qquad \wedge E \end{array} \qquad \begin{array}{l} \underline{\varphi \wedge \psi} \\ \therefore \psi \qquad \wedge E \end{array}$

Example 2.2.2. *for "p, q, r" we can prove these.*

1. $p \wedge q / \therefore q \wedge p$
2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$
3. $p \to q / \therefore (p \wedge r) \to q$
4. $(p \wedge q) \to r / \therefore p \to (q \to r)$
5. $p \to (q \to r) / \therefore (p \wedge r) \to q$

1	$p \wedge q$	ass
2	p	$\wedge E, 1$
3	q	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$

1	$(p \wedge q) \wedge r$	ass
2	$p \wedge q$	$\wedge E, 1$
3	r	$\wedge E, 1$
4	p	$\wedge E, 2$
5	q	$\wedge E, 2$
6	$q \wedge r$	$\wedge I, 3, 5$
7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \to q$	ass
2	$p \wedge r$	acp
3	p	$\wedge E, 2$
4	q	$\to E, 1, 3$
5	$(p \wedge r) \to q$	$\to I$

1	$(p \wedge q) \to r$	ass
2	p	acp
3	q	acp
4	$p \wedge q$	$\wedge I 2, 3$
5	r	$\to E 1, 4$
6	$q \to r$	$\to I$
7	$p \to (q \to r)$	$\to I$

1	$p \to (q \to r)$	ass
2	$p \wedge q$	acp
3	p	$\wedge E, 2$
4	$q \to r$	$\to E, 1, 3$
5	q	$\wedge E, 2$
6	r	$\to E, 4, 5$
7	$(p \wedge q) \to r$	$\to I$

$\dfrac{\varphi}{\therefore \varphi \vee \psi} \quad \vee I \qquad \dfrac{\varphi}{\therefore \psi \vee \varphi} \quad \vee I$	$\begin{array}{l} \varphi \vee \psi \\ \quad \left\vert \begin{array}{l} \varphi \\ \vdots \\ \theta \end{array} \right. \\ \quad \left\vert \begin{array}{l} \psi \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \qquad \vee E \end{array}$

Example 2.2.3. *for "p, q, r" we can prove these.*

1. $p \vee q / \therefore q \vee p$
2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$
3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$
4. $(p \to q) \wedge (r \to s) / \therefore (p \vee r) \to (q \vee s)$

1	$p \vee q$	ass
2	$\quad \vert\ p$	acp
3	$\quad \vert\ p \vee q$	$\vee I, 2$
4	$\quad \vert\ q$	acp
5	$\quad \vert\ q \vee p$	$\vee I, 4$
6	$q \vee p$	$\vee E, 1$

1	$(p \vee q) \vee r$	ass
2	$\quad \vert\ p \vee q$	acp
3	$\quad \vert\ \quad \vert\ p$	acp
4	$\quad \vert\ \quad \vert\ p \vee (q \vee r)$	$\vee I, 3$
5	$\quad \vert\ \quad \vert\ q$	acp
6	$\quad \vert\ \quad \vert\ q \vee r$	$\vee I, 5$
7	$\quad \vert\ \quad \vert\ p \vee (q \vee r)$	$\vee I, 6$
8	$\quad \vert\ p \vee (q \vee r)$	$\vee E, 2$
9	$\quad \vert\ r$	acp
10	$\quad \vert\ q \vee r$	$\vee I, 9$
11	$\quad \vert\ p \vee (q \vee r)$	$\vee I, 10$
12	$p \vee (q \vee r)$	$\vee E, 1$

1	$p \wedge (q \vee r)$	ass
2	p	$\wedge E1$
3	$q \vee r$	$\wedge E1$
4	q	acp
5	$p \wedge q$	$\wedge E2, 4$
6	$(p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	r	acp
8	$p \wedge r$	$I\wedge$ *2,7*
9	$(p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$(p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \to q) \wedge (r \to s)$	ass
2	$(p \to q)$	$\wedge E1$
3	$(r \to s)$	$\wedge E1$
4	$p \vee r$	acp
5	p	acp
6	q	$\to E2, 5$
7	$q \vee s$	$\vee I, 6$
8	r	acp
9	s	$\to E3, 8$
10	$q \vee s$	$\vee I9$
11	$q \vee s$	$\vee E, 4$
12	$(p \vee r) \to (q \vee s)$	$\to I$

definition 2.2.2. $\bot$ *:* $\varphi \wedge \neg\varphi$

φ $\vdots$ $\bot$ $\therefore \neg\varphi$ $\quad \neg I$	$\neg\varphi$ $\vdots$ $\bot$ $\therefore \varphi$ $\quad \neg E$

definition 2.2.3. ***schema of prove*** *you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form.you can't prove it directly. here we use a technique calls* ***"schema of prove"****. it let us prove for all situations at the same time.*

Theorem 2.2.1. (*schema*) $(\bot I, \bot E)$

φ $\neg\varphi$ ――― $\therefore \bot$ $\quad \bot I$	$\bot$ ――― $\therefore \varphi$ $\quad \bot E$

1	φ	ass
2	$\neg\varphi$	ass
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\bot$	$def.3$

1	$\bot$	ass
2	$\neg\varphi$	acp
3	$\vdots$	
4	$\bot$	$rp1$
5	φ	$\neg E$

Attention 2.2.1. *Intuitionists can skip the "$\neg E$" rule that is the "RAA" rule in old notation and use "$\bot E$" as a rule instead.*

Example 2.2.4. *for "p, q, r" we can prove these.*

1. $p \to q, \neg q / \therefore \neg p$

2. $p \to q / \therefore \neg p \lor q$

1	$p \to q$	ass
2	$\neg q$	ass
3	$\quad\mid p$	acp
4	$\quad\mid q$	$\to E, 1, 3$
5	$\quad\mid \bot$	$\bot I, 2, 4$
6	$\neg p$	$\neg I$

1	$p \to q$	ass
2	$\quad\mid \neg(\neg p \lor q)$	acp
3	$\quad\mid\quad\mid p$	acp
4	$\quad\mid\quad\mid q$	$\to E, 1, 3$
5	$\quad\mid\quad\mid \neg p \lor q$	$\lor I, 4$
6	$\quad\mid\quad\mid \bot$	$\bot I, 2, 5$
7	$\quad\mid \neg p$	$\neg I$
8	$\quad\mid \neg p \lor q$	$\lor I, 7$
9	$\quad\mid \bot$	$\bot I, 2, 8$
10	$\neg p \lor q$	$\neg I$

definition 2.2.4. $\varphi \leftrightarrow \psi$: $(\varphi \to \psi) \land (\psi \to \varphi)$

Theorem 2.2.2. (*schema*) $(\leftrightarrow I, \leftrightarrow E)$

$$\begin{array}{|l}\varphi\\ \vdots\\ \psi\end{array} \qquad \begin{array}{|l}\psi\\ \vdots\\ \varphi\end{array} \qquad \therefore \varphi \leftrightarrow \psi \quad \leftrightarrow I$$

$$\frac{\varphi \leftrightarrow \psi \quad \varphi}{\therefore \psi} \; \leftrightarrow E \qquad \frac{\varphi \leftrightarrow \psi \quad \psi}{\therefore \varphi} \; \leftrightarrow E$$

1	φ	acp
	$\vdots$	
n	ψ	
$n+1$	$\varphi \to \psi$	$\to, I$
$n+2$	ψ	acp
	$\vdots$	
m	φ	
$m+1$	$\psi \to \varphi$	$\to, I$
$m+2$	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$\wedge I$
$m+3$	$\varphi \leftrightarrow \psi$	$def., m+3$

1	$\varphi \leftrightarrow \psi$	ass
2	φ	ass
3	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$def.1$
4	$\varphi \to \psi$	$\wedge E3$
5	ψ	$\to E1, 4$

Exercise 2.2.1. *for "p, q, r" prove.*

1. $p / \therefore p$
2. $/ \therefore p \vee \neg p$
3. $\neg p, p \vee q / \therefore q$
4. $\neg p / \therefore p \to q$
5. $p \to q \therefore / \therefore \neg q \to \neg p$
6. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$
7. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$ *(De morgan)*
8. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$ *(De morgan)*
9. $(\neg p \to q) / \therefore ((p \to q) \to q)$
10. $/ \therefore (p \wedge q \to r) \wedge (p \wedge s \to r) \to (p \wedge (q \vee s) \to r)$
11. $/ \therefore (p \wedge q \to r) \vee (p \wedge s \to r) \to (p \wedge (q \wedge s) \to r)$
12. $/ \therefore (p \to (q \to r)) \to ((p \to q) \to (p \to r))$
13. $/ \therefore (p \to q) \to ((q \to r) \to (p \to r))$

Attention 2.2.2. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

This page intentionally left blank.

Chapter 3

first-order logic

3.1 what is formula?

obj are two types.we know them and we know what exactly are we talking about and unknowns are mystery.all the obj we have used until now were known ones we call them **"constants"**.

Undefined 3.1.1. ***variables*** *A variable is a symbol (usually a letter like* $x, y, or z$*) that represents an unknown obj.*

definition 3.1.1. ***Atomic formula*** *we define atomic formulas like this.*

1. *if "*x*" is an obj(constant or variable) then "*$Set(x)$*" is an atomic formula.*
2. *if "*x, y*" are obj(constant or variable) then "*$x = y$*" is an atomic formula.*
3. *if "*x, y*" are obj(constant or variable) then "*$x \in y$*" is an atomic formula.*

we denote atomic formulas with "A_i*".*

definition 3.1.2. ***well formed formula(WWF)*** *these are formulas.*

1. *atomic formulas* (A_i)*.*
2. *if* A *is a formula then* $\neg A$ *is a formula too.*
3. *if* A, B *are two formulas then* $A \bigcirc B$ *is a formula too.(*$\bigcirc$ *can be* $\rightarrow, \wedge, \vee$*).*

for example $(A_0 \rightarrow A_1, \neg A_3)$
we easily call WWF "formula" and also we denote WWF with $A, B, C, ...$

definition 3.1.3. ***open formula*** *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call **"closed formulas"** or **"sentence"**. Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.

definition 3.1.4. ***Substitution for obj*** *we define Substitution an obj with variable like this.*

1. *for variables if* $x = y$ *then* $y[a/x] := a$ *and if* $x \neq y$ *then* $y[a/x] := y$*.*

2. *for constants* $c[a/x] := c$.

definition 3.1.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like* $x = y$ *or* $x \in y$ *(*x, y *can be both constants and variables). we define it like this.*

$$Set(y)[a/x] := Set(y[a/x])$$

$$(x = y)[a/z] := (x[a/z] = y[a/z])$$

$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. *if formula* A *is complex we define it like this.*

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ *can be* $\rightarrow, \wedge, \vee$*).*

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x.we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 3.1.6. ***Open proposition*** *assume formula "A" with variables "*$x_0, x_1, .., x_n$*" in it and a set "U". we say formula "A"is a "open proposition in U" when for every obj "*$a_0, a_1, .., a_n$*" in "U"*

$$A([a_0/x_0][a_1/x_1]...[a_n/x_n])$$

is a proposition we call it open proposition and denote it with "$(P(x), Q(x, y), ..)$*"*

definition 3.1.7. ***Big "AND" and "OR"*** *assume a set like "*$\{a_0, a_1, a_2, ..., a_n\}$*" we define **big "AND"** like this.*

$$\bigwedge_{i=0}^{n} P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge ... \wedge P(a_n)$$

*and **big "OR"***

$$\bigvee_{i=0}^{n} P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee ... \vee P(a_n)$$

definition 3.1.8. ***Universe of discourse*** *we specify a set like "*$\{a_0, a_1, a_2, ..., a_n\}$*" as universe of a Logic world.*

Rule 3.1.1. ***use of open propositions*** *assume universe of discourse is "*$\{a_0, a_1, a_2, ..., a_n\}$*"*

$c \quad P(x)$

means :

$c_0 \quad P[a_0/x]$

$c_1 \quad P[a_1/x]$

$c_2 \quad P[a_2/x]$

$\vdots$

$c_n \quad P[a_n/x]$

if Px was been assemed then like this

$$\begin{array}{l|l} c & P(x) \\ & \vdots \end{array}$$

means :

$$\begin{array}{l|l} c_0 & P[a_0/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_1 & P[a_1/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_2 & P[a_2/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_n & P[a_n/x] \\ & \vdots \end{array}$$

definition 3.1.9. ***predicate schema(variable predicate)*** *a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, ...$)*

definition 3.1.10. ***Universal quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\forall x P(x)$"*

$$\forall x \phi(x) : \bigwedge_{i=0}^{n} \phi[a_i/x]$$

definition 3.1.11. ***Existential quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\exists x Px$"*

$$\exists x \phi(x) : \bigvee_{i=0}^{n} \phi[a_i/x]$$

as you can see "$\forall x \phi(x)$" and "$\exists x \phi(x)$" are sentences.

definition 3.1.12. ***Free variables*** *variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sentece is not necessary a proposition)*

3.2 Rules of first order logic.

Theorem 3.2.1. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$ *assume universe of discourse is "*$\{a_0, a_1, a_2, ..., a_n\}$*"*

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x\phi(x)} \quad \forall I$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed.*
$\dfrac{\forall x\phi(x)}{\therefore \phi[a/x]} \quad \forall E$	*1. a is an obj in universe of discourse.(it can be constant or variable.)*
$\dfrac{\phi[a/x]}{\therefore \exists x\phi(x)} \quad \exists I$	*1. a is an obj in universe of discourse.(it can be conatant or variable.)*
$\exists x\phi(x)$ | $\phi(x)$ | $\vdots$ | θ $\therefore \theta \quad \exists E$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed in previous lines.(except ones that are closed)* *3. x is not free in θ*

1	$\phi(x)$	ass
2	$\phi[a_0/x]$	$def.1$
	$\vdots$	
n	$\phi[a_n/x]$	$def.1$
$n+1$	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$\wedge I, 2, 3, ..., n+2$
$n+2$	$\forall x \phi(x)$	$def.n+3$

1	$\forall x \phi(x)$	ass
2	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$def.1$
3	$\phi[a/x]$	$\wedge E2$

1	$\phi[a/x]$	
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$	$\vee I1$
3	$\exists x \phi(x)$	$def2$

1	$\exists x \phi(x)$	
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$	
3	$\mid\ \phi[a_0/x]$	acp
	$\mid\ \vdots$	
n	$\mid\ \theta$	
$n+1$	$\mid\ \phi[a_1/x]$	acp
	$\mid\ \vdots$	
m	$\mid\ \theta$	
	$\vdots$	
l	$\mid\ \phi[a_2/x]$	acp
	$\mid\ \vdots$	
r	$\mid\ \theta$	
$r+1$	θ	$\exists E, 3$

Rule 3.2.1. (*schema*) ($I =, E =$)

$\therefore a = a \qquad I =$	$\dfrac{a = b \quad \phi(a)}{\therefore \phi(b)} \quad E = \qquad \dfrac{a = b \quad \phi(a)}{\therefore \phi(a)} \quad E =$

definition 3.2.1. ***Uniqueness*** *we define "$\exists! x \phi(x)$"*

$$\exists! x \phi(x) : \exists x(\phi(x) \wedge \forall y(\phi(y) \to x = y))$$

Example 3.2.1. *for P,Q,R we can prove these.Proof of last ones Is Left As An Exercise To The Reader.*😀

1. $\forall x(P(x) \wedge Q(x)) \therefore / \therefore \forall x P(x) \wedge \forall x Q(x)$ *(D F $\forall\wedge$)(Distribution and Factorization of $\forall\wedge$)*
2. $\exists x(P(x) \vee Q(x)) \therefore / \therefore \exists x P(x) \vee \exists x Q(x)$ *(D F $\exists\vee$)*

3. $\forall x P(x) \vee \forall x Q(x) / \therefore \forall x(P(x) \vee Q(x))$ *(F $\forall\vee$)*

4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists x P(x) \wedge \exists x Q(x)$ *(D $\exists\wedge$)*

5. $\forall x(P(x) \to Q(x)) / \therefore \forall x P(x) \to \forall x Q(x)$ *(D $\forall \to$)*

6. $\exists x P(x) \to \exists x Q(x) / \therefore \exists x(P(x) \to Q(x))$ *(F $\exists \to$)*

7. $\forall x P(x) \wedge Q \therefore / \therefore \forall x(P(x) \wedge Q)$

8. $\forall x P(x) \vee Q \therefore / \therefore \forall x(P(x) \vee Q)$

9. $\exists x P(x) \wedge Q \therefore / \therefore \exists x(P(x) \wedge Q)$

10. $\exists x P(x) \vee Q \therefore / \therefore \exists x(P(x) \vee Q)$

11. $Q \to \forall x P(x) \therefore / \therefore \forall x(Q \to P(x))$

12. $Q \to \exists x P(x) \therefore / \therefore \exists x(Q \to P(x))$

13. $\forall x P(x) \to Q \therefore / \therefore \exists x(P(x) \to Q)$

14. $\exists x P(x) \to Q \therefore / \therefore \forall x(P(x) \to Q)$

15. $\exists x \neg P(x) \therefore / \therefore \neg \forall x P(x)$ *(De morgan)*

16. $\forall x \neg P(x) \therefore / \therefore \neg \exists x P(x)$ *(De morgan)*

1	$\forall x(P(x) \wedge Q(x))$	*ass*
2	$P(x) \wedge Q(x)$	*acp*
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\forall x P(x)$	$\forall I$
6	$\forall x Q(x)$	$\forall I$
7	$\forall x P(x) \wedge \forall x Q(x)$	$\wedge I$

1	$\forall x P(x) \wedge \forall x Q(x)$	*ass*
2	$\forall x P(x)$	$\wedge E$
3	$\forall x Q(x)$	$\wedge E$
4	$P(x)$	$\forall E$
5	$Q(x)$	$\forall E$
6	$P(x) \wedge Q(x)$	$\wedge I$
7	$\forall x(P(x) \wedge Q(x))$	$\forall I$

1	$\exists x(P(x) \vee Q(x))$	ass
2	$P(x) \vee Q(x)$	acp
3	$P(x)$	acp
4	$\exists x P(x)$	$\exists I$
5	$\exists x P(x) \vee \exists x Q(x)$	$\vee I$
6	$Q(x)$	acp
7	$\exists x Q(x)$	$\exists I$
8	$\exists x P(x) \vee \exists x Q(x)$	$\vee I$
9	$\exists x P(x) \vee \exists x Q(x)$	$\vee E$
10	$\exists x P(x) \vee \exists x Q(x)$	$\exists E$

1	$\exists x P(x) \vee \exists x Q(x)$	ass
2	$\exists x P(x)$	acp
3	$P(x)$	acp
4	$P(x) \vee Q(x)$	$\vee I$
5	$\exists x(P(x) \vee Q(x))$	$\exists I$
6	$\exists x(P(x) \vee Q(x))$	$\exists E$
7	$\exists x Q(x)$	acp
8	$Q(x)$	acp
9	$P(x) \vee Q(x)$	$\vee I$
10	$\exists x(P(x) \vee Q(x))$	$\exists I$
11	$\exists x(P(x) \vee Q(x))$	$\exists E$
12	$\exists x(P(x) \vee Q(x))$	$\vee E$

1	$\forall x P(x) \vee \forall x Q(x)$	ass
2	$\forall x P(x)$	acp
3	$P(x)$	$\forall E$
4	$P(x) \vee Q(x)$	$\vee I$
5	$\forall x(P(x) \vee Q(x))$	$\forall I$
6	$\forall x Q(x)$	acp
7	$Q(x)$	$\forall E$
8	$P(x) \vee Q(x)$	$\vee I$
9	$\forall x(P(x) \vee Q(x))$	$\forall I$
10	$\forall x(P(x) \vee Q(x))$	$\vee E$

1	$\exists x(P(x) \wedge Q(x))$	ass
2	$P(x) \wedge Q(x)$	acp
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\exists x P(x)$	$\exists I$
6	$\exists x Q(x)$	$\exists I$
7	$\exists x P(x)$	$\exists E$
8	$\exists x Q(x)$	$\exists E$
9	$\exists x P(x) \wedge \exists x Q(x)$	$\wedge I$

1	$\forall x(P(x) \to Q(x))$	*ass*
2	$\forall xP(x)$	*acp*
3	$P(x) \to Q(x)$	$\forall E$
4	$Q(x)$	$\to E$
5	$\forall Q(x)$	$\forall I$
6	$\forall xP(x) \to \forall xQ(x)$	$\to I$

1	$\exists xP(x) \to \exists xQ(x)$	$\to I$
2	$\exists xP(x) \lor \neg(\exists xP(x))$	*ex.*
3	$\exists xP(x)$	*acp*
4	$\exists Q(x)$	$\to I$
5	$Q(x)$	*acp*
6	$P(x) \to Q(x)$	*ex.*
7	$\exists x(P(x) \to Q(x))$	$\exists I$
8	$\exists x(P(x) \to Q(x))$	$\exists E$
9	$\neg(\exists xP(x))$	*acp*
10	$\forall x\neg P(x)$	*De morgan*
11	$\neg P(x)$	$\forall E$
12	$P(x) \to Q(x)$	*ex.*
13	$\exists x(P(x) \to Q(x))$	$\exists I$
14	$\exists x(P(x) \to Q(x))$	$\lor E$

Exercise 3.2.1. *for P,Q,R prove.*

1. $\exists x\Big(P(x) \land \exists yQ(y) \to \forall zR(z)\Big)/ \therefore \forall y\forall z\Big(\forall xP(x) \land Q(y) \to R(z)\Big)$
2. $\forall x\exists y\Big(P(x) \land Q(y) \to \forall zR(z)\Big)/ \therefore \forall z\Big(\exists xP(x) \land \forall yQ(y) \to R(z)\Big)$
3. $\forall xP(x) \to \exists yQ(y)/ \therefore \exists x\exists y\Big(P(x) \to Q(y)\Big)$
4. $\forall x\forall z\exists y\exists w\Big(P(y,z) \lor Q(z) \to R(x,w)\Big)/ \therefore \Big(\exists z\forall yP(y,z) \lor \exists zQ(z)\Big) \to \forall x\exists wR(x,w)$
5. $\exists xP(x) \lor \exists x\Big(Q(x) \land R(x)\Big)/ \therefore \exists x\Big(P(x) \lor Q(x)\Big) \land \exists x\Big(P(x) \lor R(x)\Big)$
6. $\exists x\exists z\forall y\Big(P(x,y) \to Q(z) \lor R(x,y)\Big)/ \therefore \forall x\exists yP(x,y) \to \Big(\exists zQ(z) \lor \exists x\exists zR(x,y)\Big)$
7. $\forall x\exists y\Big(\exists zP(x,y,z) \land Q(x,y)\Big) \lor \forall x\exists y\exists z\Big(P(x,y,z) \land R(x,y)\Big)/ \therefore \forall x\exists y\exists z\Big(P(x,y,z) \land (Q(x,y) \lor R(x,y))\Big)$

Attention 3.2.1. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 4

The beginning of set theory

In language of set theory if we are talking about "$\forall X(P(X))$" it means "$\forall x(Set(x) \to P(x))$" and when we are talking about "$\exists X(P(X))$" it means "$\exists x(Set(x) \wedge P(x))$". similarly if we are talking about "$\forall x \in A \quad P(x)$" it means "$\forall x(x \in A \to P(x))$" and when we are talking about "$\exists x \in A \quad P(x)$" it means "$\exists x(x \in A \wedge P(x))$".

4.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 4.1.1. ***Existence of universe of set theory*** *There exists a set V (we denote it as V.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition "ϕ" in "set theory" we mean proposition "ϕ" in "V").*

definition 4.1.1. ***Subset*** *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \to x \in B)$$

or i.e.

$$\forall x \in A \quad x \in B$$

Theorem 4.1.1. $(schema)$ $(\subseteq I, \subseteq E)$

$\begin{array}{\|l} x \in A \\ \vdots \\ x \in B \end{array}$ $\therefore A \subseteq B \qquad \subseteq I$	$A \subseteq B$ $\underline{a \in A}$ $\therefore a \in B \qquad \subseteq E$

1	$\quad\mid x \in A$	*acp*
	$\quad\mid \vdots$	
n	$\quad\mid x \in B$	
$n+1$	$x \in A \to x \in B$	$I \to, 1, n$
$n+2$	$\forall x(x \in A \to x \in B)$	$\forall I, n+1$
$n+3$	$A \subseteq B$	$def.n+2$

1	$A \subseteq B$	*ass*
2	$a \in A$	*ass*
3	$\forall x(x \in A \to x \in A)$	$def.1$
4	$a \in A \to a \in B$	$\forall E, 4$
5	$a \in B$	$E \to 3, 5$

Example 4.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	*ass*
2	$B \subseteq C$	*ass*
3	$\quad\mid x \in A$	*acp*
4	$\quad\mid x \in B$	$\subseteq E, 1, 3$
5	$\quad\mid x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 4.1.2. ***Extensionality*** *Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \to X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$$

Theorem 4.1.2. (*schema*) $(I =, E =)$

$\mid x \in A$ $\mid \vdots$ $\mid x \in B$ $\mid x \in B$ $\mid \vdots$ $\mid x \in A$ $\therefore A = B \qquad I =$	$\dfrac{\begin{array}{c}A = B\\ a \in A\end{array}}{\therefore a \in B} \qquad E =$	$\dfrac{\begin{array}{c}A = B\\ a \in B\end{array}}{\therefore a \in A} \qquad E =$

1	$\quad x \in A$	*acp*
	$\quad \vdots$	
n	$\quad x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$\quad x \in B$	*acp*
	$\quad \vdots$	
m	$\quad x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$	*Extensionality*
$m+3$	$(A \subseteq B \wedge B \subseteq A) \to A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \to, m+3, m+4$

Axiom 4.1.3. ***Weak comprehension*** (*schema*) *this axioms says for every open proposition like "$\phi(x, x_0, x_1, ..., x_n)$" there is a set like "$X$" such that every "$x$" is a member of that set if and only if it holds "$\phi(x, x_0, x_1, ..., x_n)$".*

$$\forall x_0, x_1, ..., x_n \exists X \forall x(\phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as "$\{x \mid \phi(x, x_0, x_1, ..., x_n)\}$". sometimes instead of "$\{x \mid x \in A \wedge \phi(x, x_0, x_1, ..., x_n)\}$" we write "$\{x \in A \mid \phi(x, x_0, x_1, ..., x_n)\}$".

this axiom show that why Russell's paradox doesn't happened here because if ou assume "$R = \{x | x \notin x\}$" is a set. you can't prove or disprove "$R \in R$" so "$x \notin x$" is not a proposition in "set theorhy". (we don't need to prove that now.we will prove it in"'**MetaLogic'**"). (it's not obvios in "$x \notin x$" witch set holds and witch not). (and likely "$\{X \mid a \in X\}$" is not a set either).

4.2 definitions of set theory

Let's talk about some definitions in set theoty.These definitions will come in handy later.

definition 4.2.1. ***Empty set*** *also known as the null set, is the unique mathematical set containing no elements at all. (uniqueness can be proven.)*

$$\varnothing := \{x \mid \bot\}$$

therefore.

$$\forall x(x \in \varnothing \leftrightarrow \bot)$$

therefore.

$$\forall x(x \notin \varnothing)$$

Example 4.2.1. $/ \therefore \forall X(\varnothing \subseteq X)$

1	$\quad\vert\; x \in \varnothing$	*acp*
2	$\quad\vert\; \forall x(x \notin \varnothing)$	*def*
3	$\quad\vert\; x \notin \varnothing$	$\forall E, 2$
4	$\quad\vert\; \bot$	$\bot I, 1, 3$
5	$\quad\vert\; x \in X$	$\bot E, 4$
6	$\varnothing \subseteq X$	$\subseteq I, 1, 5$
7	$\forall X(\varnothing \subseteq X)$	$\forall I, 6$

Lemma 4.2.1. $A = \varnothing \therefore / \therefore \forall y(y \notin A)$

1	$\forall y(y \notin A)$	*ass*
2	$\quad\vert\; y \in A$	*acp*
3	$\quad\vert\; y \notin A$	$\forall E, 1$
4	$\quad\vert\; \bot$	$\bot I, 2, 3$
5	$\quad\vert\; y \in \varnothing$	$\bot E, 4$
6	$\quad\vert\; y \in \varnothing$	*acp*
7	$\quad\vert\; \forall y(y \notin \varnothing)$	*def.*
8	$\quad\vert\; y \notin \varnothing$	$\forall E, 7$
9	$\quad\vert\; \bot$	$\bot I, 6, 8$
10	$\quad\vert\; y \in A$	$\bot E, 9$
11	$A = \varnothing$	$I =$

1	$A = \varnothing$	*ass*
2	$\forall y(y \notin \varnothing)$	*def.*
3	$\forall y(y \notin A)$	$= E, 1, 2$

Theorem 4.2.1. (*schema*) $(I\varnothing \neq, E\varnothing \neq, \varnothing = I, \varnothing = E)$

$\dfrac{a \in A}{\therefore A \neq \varnothing}$ $\quad I\varnothing \neq$	$A \neq \varnothing$ $\quad\vert\; x \in A$ $\quad\vert\; \vdots$ $\quad\vert\; \theta$ $\therefore \theta \qquad E\varnothing \neq$

$\begin{array}{l} \quad \left\vert \begin{array}{l} x \in A \\ \vdots \\ \bot \end{array} \right. \\ \therefore A = \varnothing \qquad \varnothing = I \end{array}$	$\dfrac{A = \varnothing}{\therefore a \notin A} \qquad \varnothing = E$

1	$a \in A$	*ass*
2	$\exists x(x \in A)$	$\exists I, 1$
3	$\neg(\forall x(x \notin A))$	*De Morgan* 2
4	$A \neq \varnothing$	*lemma* 4.4.1, 3

1	$\quad\vert\ x \in A$	*acp*
	$\quad\vert\ \vdots$	
n	$\quad\vert\ \bot$	
$n+1$	$x \notin A$	$\neg I$
$n+2$	$\forall x(x \notin A)$	$\forall I, n+1$
$n+3$	$A = \varnothing$	*lemma* 4.4.1, 1

1	$A = \varnothing$	*ass*
2	$\forall x(x \notin A)$	*lemma* 4.4.1, 1
3	$a \notin A$	$\forall E2$

definition 4.2.2. ***pairing set***

$$\{a_0, a_1, a_2, ..., a_n\} := \{x \mid x = a_0 \lor x = a_1 \lor x = a_2 \lor ... \lor x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, ..., a_n\} \leftrightarrow (x = a_0 \lor x = a_1 \lor x = a_2 \lor ... \lor x = a_n))$$

Theorem 4.2.2. (*schema*) $(I\{\}, E\{\})$

$\dfrac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, ..., a_n\}} \qquad I\{\}$	$\begin{array}{l} a \in \{a_0, a_1, a_2, ..., a_n\} \\ \quad \left\vert \begin{array}{l} a = a_0 \\ \vdots \\ \theta \end{array} \right. \\ \vdots \\ \quad \left\vert \begin{array}{l} a = a_n \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \qquad\qquad E\{\} \end{array}$

Proof Is Left As An Exercise To The Reader.😃

definition 4.2.3. ***Union and Intersection***
Intersection:

$$A \cap B := \{x \mid x \in A \wedge x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \wedge x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \vee x \in B\}$$

therefore.

$$\forall x(x \in A \cup B \leftrightarrow (x \in A \vee x \in B))$$

Theorem 4.2.3. *(schema)* $(\cap I, \cap E, \cup I, \cup E)$

<table>
<tr>
<td>$\begin{array}{l} a \in A \\ a \in B \\ \hline \therefore a \in A \cap B \end{array} \qquad \cap I$</td>
<td>$\begin{array}{l} a \in A \cap B \\ \hline \therefore a \in A \end{array} \qquad \cap E \qquad \begin{array}{l} a \in A \cap B \\ \hline \therefore a \in B \end{array} \qquad \cap E$</td>
</tr>
<tr>
<td>$\begin{array}{l} a \in A \\ \hline \therefore a \in A \cup B \end{array} \qquad \cup I$
$\begin{array}{l} a \in A \\ \hline \therefore a \in B \cup A \end{array} \qquad \cup I$</td>
<td>$\begin{array}{l} a \in A \cup B \\ \quad \begin{array}{|l} a \in A \\ \vdots \\ \theta \end{array} \\ \quad \begin{array}{|l} a \in B \\ \vdots \\ \theta \end{array} \\ \therefore \theta \qquad\qquad \cup E \end{array}$</td>
</tr>
</table>

Proof Is Left As An Exercise To The Reader.😃

somtimes all members of a set is also set we call it “set of sets”.

definition 4.2.4. ***Big Union and Intersection*** *if “A” is “set of sets” and “$A \neq \varnothing$” we define Big Intersection:*

$$\bigcap A := \{x \mid \forall X(X \in A \rightarrow x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcap A \leftrightarrow \forall X(X \in A \rightarrow x \in X)\Big)$$

Big Union:

$$\bigcup A := \{x \mid \exists X(X \in A \land x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x(x \in \bigcup A \leftrightarrow (\exists X(X \in A \land x \in X)))$$

Theorem 4.2.4. (*schema*) ($\bigcap I, \bigcap E, \bigcup I, \bigcup E$)

$\begin{array}{l\|l} & X \in A \\ & \vdots \\ & a \in X \end{array}$ $\therefore a \in \bigcap A \qquad \bigcap I$	$a \in \bigcap A$ $B \in A$ $\therefore a \in B \qquad \bigcap E$
$a \in B$ $B \in A$ $\therefore a \in \bigcup A \qquad \bigcup I$	$a \in \bigcup A$ $\begin{array}{l\|l} & a \in X \\ & X \in A \\ & \vdots \\ & \theta \end{array}$ $\therefore \theta \qquad \bigcup E$

Proof Is Left As An Exercise To The Reader.😀

definition 4.2.5. ***Complement***

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow x \notin A)$$

Theorem 4.2.5. (*schema*) ($()^c I, ()^c E$)

$\begin{array}{l\|l} & a \in A \\ & \vdots \\ & \bot \end{array}$ $\therefore a \notin A \qquad ()^c I$	$\begin{array}{l\|l} & a \notin A \\ & \vdots \\ & \bot \end{array}$ $\therefore a \in A \qquad ()^c E$

Proof Is Left As An Exercise To The Reader.😀

Observation 4.2.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow \top)$$

therefore.

$$\forall x(x \in V)$$

definition 4.2.6. ***Subtraction***

$$A - B := \{x \mid x \in A \wedge x \notin B\}$$

therefore.

$$\forall x\Big(x \in A - B \leftrightarrow (x \in A \wedge x \notin B)\Big)$$

Theorem 4.2.6. *(schema)* $(-I, -E)$

$a \in A$ $a \notin B$ $\therefore a \in A - B \quad -I$	$\dfrac{a \in A - B}{\therefore a \in A} \quad -E \qquad \dfrac{a \in A - B}{\therefore a \notin B} \quad -E$

Proof Is Left As An Exercise To The Reader.😃

Lemma 4.2.2. $/ \therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader.😃

definition 4.2.7. ***Powerset***

$$P(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X\Big(X \in P(A) \leftrightarrow (X \subseteq A)\Big)$$

Theorem 4.2.7. *(schema)* (PI, PE)

$\dfrac{A \subseteq B}{\therefore A \in P(B)} \quad PI$	$\dfrac{A \in P(B)}{\therefore A \subseteq B} \quad PE$

Proof Is Left As An Exercise To The Reader.😃

from now on we don't get in to all details of proofs because it will be too long and hard to read. so we will take it easy from now and don't write all details like reasons of lines and all lines we left it to reader to undrestand what happened there.😃

Chapter 5

Relations and Functions

5.1 Orderd pair and Cartesian product

definition 5.1.1. ***Ordered pair***

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

definition 5.1.2. ***Tuples****(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)*

$$(x_1, x_2, ..., x_n) := ((x_1, x_2, ..., x_{n-1}), x_n)$$

Lemma 5.1.1. $/ \therefore (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader.😀

definition 5.1.3. ***Cartesian product***

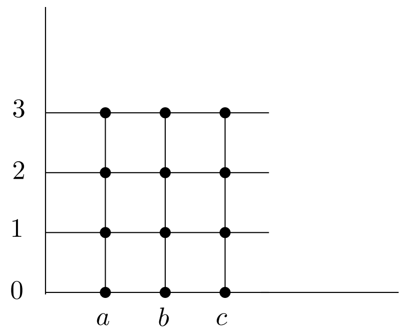
$$A \times B := \{u \mid \exists x, y\Big((x, y) = u \wedge x \in A \wedge y \in B\Big)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 5.1.1. *if* $A = \{a, b, c\}$ *and* $B = \{0, 1, 2, 3\}$ *then*

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



Exercise 5.1.1. *for sets A, B, C, prove.*

1. $/ \therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$

2. $/ \therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$

3. $/ \therefore A \times (B - C) = (A \times B) - (A \times C)$

5.2 Relations

definition 5.2.1. ***Relation*** *we call a set like* ***R*** *a* ***Relation*** *when there are two sets like X and Y s.t. $R \subseteq X \times Y$.*

$$Rel(R) : \exists X, Y(R \subseteq X \times Y).$$

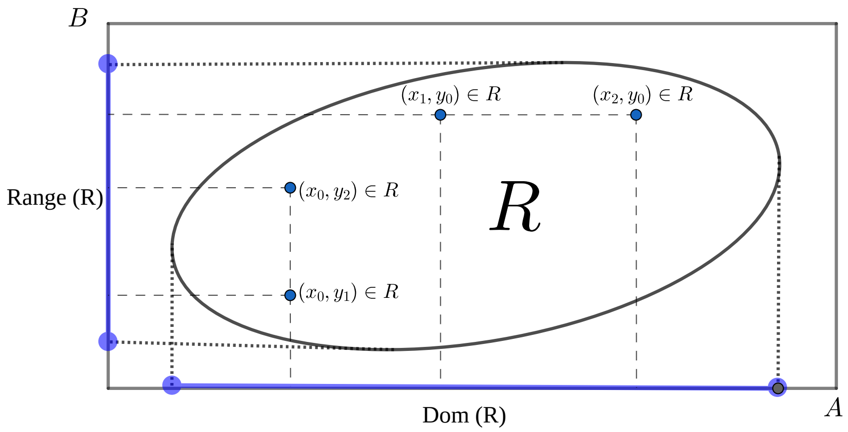
somtimes we use different notations in this book. you have to be smart and recognize it by yourself. sometimes instead of "$x \in R$" we denote it by "$R(x)$". and instead of "$(x_0, x_1, ..., x_n) \in R$" we denote it by "$R(x_0, x_1, ..., x_n)$". if tuple is binary, instead of "$(x, y) \in R$" we denote it by "xRy". or "$R(x, y)$" instead of "$R\big((x, y)\big)$"
so if we have a definition like "$R(x) : \phi(x)$" s.t. "$\phi(x)$" is a first order proposition. "weak comprehension" show us there is a straight relatioon between "propositions" and "sets" so you can see it as "$R := \{x \mid \phi(x)\}$". and somtimes we might show a formula in natural language for example below instead of "$Rel(X)$" we can say "X is relation". if it is confusing so forget it.😃

definition 5.2.2. ***Domain and Range(also called Image)***
Domain of a set "R" is all "x" that there exists a "y" s.t. "xRy"

$$Dom(R) := \{x \mid \exists y \quad xRy\}$$

Range(Image) of a set "R" is all "y" that there exists a "x" s.t. "xRy"

$$Range(R) := \{y \mid \exists x \quad xRy\}.$$



definition 5.2.3. ***Inverse of a Relation***

$$R^{-1} = \{u \mid \exists x, y\Big(u = (x, y) \wedge yRx\Big)\}$$

or i.e.

$$R^{-1} = \{(x, y) \mid yRx\}$$

therefore.

$$\forall x, y\Big(xR^{-1}y \leftrightarrow yRx\Big)$$

definition 5.2.4. ***Combination of a Relations***

$$S \circ R = \{u \mid \exists x, y\Big(u = (x, y) \wedge \exists z\big(xRz \wedge zSy\big)\Big)\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z\Big(xRz \wedge zSy\Big)\}$$

therefore.

$$\forall x, y\Big((x, y) \in S \circ R \leftrightarrow \exists z\big(xRz \wedge zSy\big)\Big)$$

definition 5.2.5. ***Image***

$$R[A] := \{y \mid \exists x \in A \quad xRy\}$$

you can olso see by definition of "inverse of a Relation" and "image" this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad xRy\}$$

*we cal this "**Pre-image**"*

5.3 Functions

definition 5.3.1. ***Function*** *we call a relation "**f**" a "**function**" when every input has exactly one output i.e.*

$$Func(f) : Rel(f) \wedge \forall x, y_1, y_2 \Big((x, y_1) \in f \wedge (x, y_2) \in f \to y_1 = y_2 \Big)$$

definition 5.3.2. ***Restriction of a function*** *to a set "X" (usually a subset of "dom(f)") is the function*

$$f \restriction X := \{(x, y) \in f \mid x \in X\}$$

definition 5.3.3. ***Output of a function*** *assume a function **f** and $(x, y) \in f$, we define "$f(x) := y$" and we call "$f(x)$" a "**output of a function**". and the reverse is also true, if we use "$f(x)$", we have a function "**f**" and we can say "$x \in Dom(f)$". "output of a function" is object too, so we can have $f(f(x))$ as well.*

Attention 5.3.1. *input of a function can be a ordered pair itself, so we can have $f((x, y))$ and it is a "**output of a function**". we define "$f(x, y) := f((x, y))$" for simplicity. this is true for tuples as well, so we can have "$f((x_1, x_2, ..., x_n))$" and we define "$f(x_1, x_2, ..., x_n) := f((x_1, x_2, ..., x_n))$" for simplicity.*

definition 5.3.4. ***Term*** *we define term like this.*

1. *every obj is a term.*
2. *if $t_0, ...t_n$ are terms and "$Func(f)$" and "$(t_0, ...t_n) \in Dom(f)$". then "$f(t_0, ...t_n)$" is also a term.*

we denote terms with small letters like $t, s, r, ...$.

definition 5.3.5. ***Substitution for term*** *we define Substitution an term with variable like this.*

1. *for variables if $x = y$ then $y[t/x] := t$ and if $x \neq y$ then $y[t/x] := y$.*
2. *for constants $c[t/x] := c$.*
3. *for terms like $f(t_0, ...t_n)$*

$$f(t_0, ...t_n)[t/x] := f(t_0[t/x], ...t_n[t/x])$$

definition 5.3.6. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single term we do it as we explain in "**Substitution for term**". if it's like $t_0 = t_1$ or $t_0 \in t_1$ (t_0, t_1 are terms). we define it like this.*

$$Set(y)[t/x] := Set(y[t/x])$$

$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$

$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[t/x] := (B[t/x] \bigcirc C[t/x])$$

($\bigcirc$ can be $\to, \wedge, \vee$).

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 5.3.1. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x \phi(x)}$ $\forall I$	*1. x is a variable.* *2. "$\phi(x)$" must not be assumed.*
$\dfrac{\forall x \phi(x)}{\therefore \phi[t/x]}$ $\forall E$	*1. t is a term.*
$\dfrac{\phi[t/x]}{\therefore \exists x \phi(x)}$ $\exists I$	*1. t is a term.*
$\exists x \phi(x)$ $\mid \phi(x)$ $\mid \vdots$ $\mid \theta$ $\therefore \theta$ $\exists E$	*1. x is a variable.* *2. "$\phi(x)$" must not be assumed in previous lines.(except ones that are closed)* *3. x is not free in θ*

$(\forall I, \exists E)$ is the same as $(\forall I, \exists E)$ in chapter three so we have to prove $(\forall E, \exists I)$. so assume $t = f(x)$ so to it's easy to see $Func(f)$ and $x \in Dom(f)$.

1	$\forall x \phi(x)$	*ass*
2	$Func(f)$	*ass*
3	$\exists y\big((x, y) \in f\big)$	*ass*
4	$\mid (x, y) \in f$	*acp*
5	$\mid f(x) = y$	*def*
6	$\mid \phi(y)$	$\forall E$
7	$\mid \phi(f(x))$	$= E$
8	$\phi(f(x))$	$\exists E$

1	$\phi(f(x))$	*ass*
2	$Func(f)$	*ass*
3	$\exists y\big((x, y) \in f\big)$	*ass*
4	$\mid (x, y) \in f$	*acp*
5	$\mid f(x) = y$	*def*
6	$\mid \phi(y)$	$= E$
7	$\mid \exists x \phi(x)$	$\exists I$
8	$\exists x \phi(x)$	$\exists E$

Attention 5.3.2. *in all rules and theorems you can replace all constants like $a, b, c, \ldots$ with terms like $t, s, r, \ldots.$ and rewrite them.*

definition 5.3.7. ***Compatible Function***

1. *Compatible*$(f, g) : \forall x \in dom(f) \cap dom(g) \quad f(x) = g(x)$.
2. *Compatible*$(F) : \forall f, g \in F \quad Compatible(f, g)$.

Lemma 5.3.1. .

1. *Compatible*$(f, g) \therefore / \therefore Func(f \cup g)$.
2. *Compatible*$(f, g) \therefore / \therefore f \restriction (dom(f) \cap dom(g)) = g \restriction (dom(f) \cap dom(g))$.

Proof Is Left As An Exercise To The Reader.😀

Theorem 5.3.2. *Compatible*$(F)/ \therefore (1), (2), (3)$

1. $Func(\bigcup F)$.
2. $dom(\bigcup F) = \bigcup\{dom(f) \mid f \in F\}$.
3. $\forall f \in F \quad Compatible(f, \bigcup F)$.

Proof Is Left As An Exercise To The Reader.😀

definition 5.3.8. ***Function from set to set*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is subset of **B** i.e.*

$$(f : A \to B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 5.3.9. ***Set of functions from set to set*** *we denote set of all "$f : A \to B$" with "B^A" i.e.*

$$B^A : \{f \mid f : A \to B\}$$

definition 5.3.10. ***Injective function*** *we call a function **f** injective when every output has at most one input i.e.*

$$Injective(f) : \forall y, x_1, x_2 \Big((x_1, y) \in f \wedge (x_2, y) \in f \to x_1 = x_2\Big).$$

*we call a function **f** injective from **A** to **B** when it's function from **A** to **B** and it's injective i.e.*

$$(f : A \xrightarrow{inj} B) : (f : A \to B) \wedge Injective(f)$$

definition 5.3.11. ***Surjective function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 5.3.12. ***Bijection function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{inj} B) : (f : A \xrightarrow{inj} B) \wedge (f : A \xrightarrow[surj]{} B)$$

5.4 Family of sets

Attention 5.4.1. ***Indexes*** *are objects (like numbers) allows us to give a unique "tag" to every object. we use* $i, j, k, ...$ *for variable indexes and we use* $m, n, l, ...$ *for constant indexes.*

definition 5.4.1. ***Family of sets*** *assume* $A : I \to \mathcal{F}$ *is a function from a nonempty set of indices* I *to a family of sets* $\mathcal{F}$*, then we call* $(Range[A])$ *an indexed family of sets. we can write* A_i *instead of* $A(i)$ *and we can write* $(A_i)_{i\in I}$ *instead of "*A*" and* $\{A_i\}_{i\in I}$ *instead of "*$Range[A]$*" or "*$A[I]$*".*

$$(A_i)_{i\in I} := A$$

and

$$\{A_i\}_{i\in I} := A[I]$$

definition 5.4.2. ***Intersrction and union of an family of sets***

$$\bigcap_{i\in I} A_i := \bigcap \{A_i\}_{i\in I}$$

$$\bigcup_{i\in I} A_i := \bigcup \{A_i\}_{i\in I}$$

Attention 5.4.2. *we can write* $\forall i \in I \quad P(i)$ *instead of* $\forall i(i \in I \to P(i))$ *and we can write* $\exists i \in I \quad P(i)$ *instead of* $\exists i(i \in I \land P(i))$*.*

Lemma 5.4.1.

$$/ \therefore \bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i\in I} A_i$	*acp*
2	$x \in \bigcap\{A_i\}_{i\in I}$	*def*.1
3	$x \in \bigcap A[I]$	*def*.1
4	$\forall Y \in A[I] \quad x \in Y$	$def \bigcap$
5	$\forall Y((\exists i \in I \quad A_i = Y) \to x \in Y)$	*defimage*
6	$(\exists i \in I \quad A_i = A_{i'}) \to x \in A_{i'}$	$\forall E$
7	$i' \in I$	*acp*
8	$A_{i'} = A_{i'}$	$= I$
9	$i' \in I \wedge A_{i'} = A_{i'}$	$\wedge I$
10	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
11	$x \in A_{i'}$	$\to E$
12	$i' \in I \to x \in A_{i'}$	$\to I$
13	$\forall i \in I \quad x \in A_i$	$\forall I$
14	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	*def*
15	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	*acp*
16	$\forall i \in I \quad x \in A_i$	*def*
17	$\exists i \in I \quad A_i = Y$	*acp*
18	$i \in I$	*acp*
19	$x \in A_i$	$\forall E$
20	$A_i = Y$	$\wedge E$
21	$x \in Y$	$= E$
22	$x \in Y$	$\exists E$
23	$(\exists i \in I \quad A_i = Y) \to x \in Y$	$\to I$
24	$\forall Y((\exists i \in I \quad A_i = Y) \to x \in Y)$	$\forall I$
25	$\forall Y \in A[I] \quad x \in Y$	*def*
26	$x \in \bigcap A[I]$	*def*
27	$x \in \bigcap\{A_i\}_{i\in I}$	*def*
28	$x \in \bigcap_{i\in I} A_i$	*def*
29	$\bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/ \therefore \bigcup_{i\in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$x \in \bigcup_{i \in I} A_i$	*acp*
2	$x \in \bigcup \{A_i\}_{i \in I}$	*def.*1
3	$x \in \bigcup A[I]$	*def.*1
4	$\exists Y \in A[I] \quad x \in Y$	*def* $\bigcup$
5	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	*def.image*
6	$x \in Y$	*acp*
7	$\exists i \in I \quad A_i = Y$	*acp*
8	$i \in I$	*acp*
9	$A_i = Y$	*acp*
10	$x \in A_i$	$= E$
11	$\exists i \in I \quad x \in A_i$	$\exists I$
12	$\exists i \in I \quad x \in A_i$	$\exists E$
13	$\exists i \in I \quad x \in A_i$	$\exists E$
14	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	*def*
15	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	*acp*
16	$\exists i \in I \quad x \in A_i$	*def*
17	$i' \in I$	*acp*
18	$x \in A_{i'}$	*acp*
19	$A_{i'} = A_{i'}$	$= I$
20	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
21	$(\exists i \in I \quad A_i = A_{i'}) \wedge x \in A_{i'}$	$\wedge I$
22	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists I$
23	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists E$
24	$\exists Y \in A[I] \quad x \in Y$	*def*
25	$x \in \bigcup A[I]$	*def*
26	$x \in \bigcup \{A_i\}_{i \in I}$	*def*
27	$x \in \bigcup_{i \in I} A_i$	*def*
28	$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$= I$

Theorem 5.4.1. (*schema*) ($\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E$)

$\left\vert \begin{array}{l} i \in I \\ \vdots \\ t \in A_i \end{array} \right.$ $\therefore t \in \bigcap_{i \in I} A_i \qquad \bigcap_I I$	$t \in \bigcap_{i \in I} A_i$ $\underline{n \in I}$ $\therefore t \in A_n \qquad \bigcap_I E$
$t \in A_n$ $\underline{n \in I}$ $\therefore t \in \bigcup_{i \in I} A_i \qquad \bigcup_I I$	$t \in \bigcup_{i \in I} A_i$ $\left\vert \begin{array}{l} i \in I \\ t \in A_i \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \qquad \bigcup_I E$

Proof Is Left As An Exercise To The Reader.😃

5.5 Structures and morphisms

definition 5.5.1. ***Structure*** *"$\mathfrak{M}$" is a tuple defined like this.*

$$\mathfrak{M} := (M, f_0, f_1, ..., f_n, R_0, R_1, ..., R_m, c_0, c_1, ..., c_k)$$

"M" is a ***nonempty*** *set shows our universe "f_i" are funcions and "R_I" are relations and "c_i" are constants. and for the case we have arbitary family of functions and relations and constants.*

$$\mathfrak{M} := (M, \{f_i\}_{i \in I_f}, \{R_i\}_{i \in I_R}, \{c_i\}_{i \in I})$$

definition 5.5.2. ***Homorphism*** *if "$\mathfrak{M} := (M, \{f_{i,m}\}_{i \in I_f}, \{R_{i,m}\}_{i \in I_R}, \{c_{i,m}\}_{i \in I})$" and "$\mathfrak{N} := (N, \{f_{i,n}\}_{i \in I_f}, \{R_{i,n}\}_{i \in I_R}, \{c_{i,n}\}_{i \in I})$" are two structures. function "$h : M \to N$"is called "(weak) Homorphism" when this conditions holds*

1. *for eveery relation "$R_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad R_{i,m}(x_0, x_1, x_2, ..., x_k) \to R_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

2. *for eveery function "$f_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad h(f_{i,m}(x_0, x_1, x_2, ..., x_k)) = f_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

3. *for eveery constant "$c_{i,m}$"*

$$h(c_{i,m}) = c_{i,n}$$

.

in condition (1) if "$\leftrightarrow$" be instead of "$\rightarrow$" then it's called "(strong) Homorphism"

few notices: when we are talking about "Homorphism" we usually talk about "(strong) Homorphism" .
in the case (in definition "(strong) Homorphism") we denote it by "$Hom(h, \mathfrak{M}, \mathfrak{N})$". if exists function "$h$" "$Hom(h, \mathfrak{M}, \mathfrak{N})$" we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are homomorphic. we denote it by "$Hom(\mathfrak{M}, \mathfrak{N})$".
if "$h : M \xrightarrow{inj} N$" we call it "Monomorphism" or "Embedding".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are momomorphic.
if "$h : M \xrightarrow[surj]{} N$" we call it "Epimorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Eomomorphic.
if "$h : M \xrightarrow[surj]{inj} N$" we call it "Isomorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Iomomorphic. and denote it by "$\mathfrak{M} \cong \mathfrak{N}$".
if "$k : M \rightarrow M$" is a function and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Indomorphism".
if "$k : M \xrightarrow[surj]{inj} M$" is a bijection and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Automorphism".

5.6 Equivalences and Partitions

definition 5.6.1. ***Equivalence relation*** *assume a structure "$\mathfrak{A} := (A, \sim)$"("$\sim$" is a relation). we say "$\sim$" is a "Equivalence relation" in "$\mathfrak{A}$" if it's reflexive,symmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \sim x$$

$$(symmetric) : \forall x, y \in A \quad x \sim y \rightarrow y \sim x$$

$$(transitive) : \forall x, y, z \in A \quad x \sim y \wedge y \sim z \rightarrow x \sim z$$

Lemma 5.6.1. *"$\cong$" is a equivalence relation in "$(Structure, \cong)$"*
Proof Is Left As An Exercise To The Reader.😃

definition 5.6.2. ***Equivalence class of "x"*** *Let "$\sim$" be an equivalence relation on "$\mathfrak{A}$". For every "$x \in A$"*

$$[x]_\sim := \{y \in A \mid y \sim x\}$$

you can also see it as a function $[\_]_\sim : A \rightarrow V$.

Lemma 5.6.2. *For every "$a, b \in A$"*

1. $a \sim b \therefore / \therefore [a]_\sim = [b]_\sim$
2. $a \nsim b \therefore / \therefore [a]_\sim \cap [b]_\sim = \varnothing$

prove.1

$/ \therefore$

$$
\begin{array}{lll}
1 & a \sim b & \\
2 & \quad \mid & x \in [a]_\sim \\
3 & \quad \mid & x \sim a \\
4 & \quad \mid & a \sim b \\
5 & \quad \mid & x \sim b \\
6 & \quad \mid & x \in [b]_\sim \\
7 & \quad \mid & x \in [b]_\sim \\
8 & \quad \mid & x \sim b \\
9 & \quad \mid & a \sim b \\
10 & \quad \mid & x \sim a \\
11 & \quad \mid & x \in [a]_\sim \\
12 & [a]_\sim = [b]_\sim &
\end{array}
$$

$\therefore /$

$$
\begin{array}{ll}
1 & [a]_\sim = [b]_\sim \\
2 & a \sim a \\
3 & a \in [a]_\sim \\
4 & a \in [b]_\sim \\
5 & a \sim b
\end{array}
$$

prove.2

$/ \therefore$

$$
\begin{array}{llll}
1 & a \nsim b & & \\
2 & \quad \mid & [a]_\sim \cap [b]_\sim \neq \varnothing & \\
3 & \quad \mid & \exists x \quad x \in [a]_\sim \cap [b]_\sim & \\
4 & \quad \mid & \quad \mid & x \in [a]_\sim \cap [b]_\sim \\
5 & \quad \mid & \quad \mid & x \sim a \\
6 & \quad \mid & \quad \mid & x \sim b \\
7 & \quad \mid & \quad \mid & a \sim b \\
8 & \quad \mid & \quad \mid & \perp \\
9 & \quad \mid & \perp & \\
10 & [a]_\sim \cap [b]_\sim = \varnothing & &
\end{array}
$$

$\therefore /$

$$
\begin{array}{lll}
1 & [a]_\sim \cap [b]_\sim = \varnothing & \\
2 & \quad \mid & a \sim b \\
3 & \quad \mid & a \sim a \\
4 & \quad \mid & a \in [a]_\sim \\
5 & \quad \mid & a \in [b]_\sim \\
6 & \quad \mid & a \in [a]_\sim \cap [b]_\sim \\
7 & \quad \mid & \perp \\
8 & a \nsim b &
\end{array}
$$

definition 5.6.3. ***Partition*** *is a "set of sets" like "P" of a set like "A" with this conditions.*

1. $\forall X \in P \quad X \neq \varnothing$
2. $\forall X, Y \in P \quad X \neq Y \rightarrow X \cap Y = \varnothing$
3. $\bigcup P = A$
   if these condition holds then we say "$Part(P, A)$".

definition 5.6.4. ***Set of Equivalence classes***

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

Theorem 5.6.1. *if "$\sim$" is a equivalence class in "$(A, \sim)$" then "$A/\sim$" is a partition for "A" i.e. "$Part(A/\sim, A)$".*

1 $[x]_\sim \in A$

2 $x \in [x]_\sim$

3 $[x]_\sim \neq \varnothing$

1 $[a]_\sim \neq [b]_\sim$

2 $a \nsim b$

3 $[a]_\sim \cap [b]_\sim = \varnothing$

1 | $x \in \bigcup(A/\sim)$

2 | | $y \in A$

3 | | $x \in [y]_\sim$

4 | | $x \in A$

5 | $x \in A$

6 | $x \in A$

7 | $x \in [x]_\sim \wedge [x]_\sim \in (A/\sim)$

8 | $\exists Y(x \in Y \wedge Y \in (A/\sim))$

9 | $x \in \bigcup(A/\sim)$

10 $A = \bigcup(A/\sim)$

Theorem 5.6.2. *if "P" is a partition for "A" we define*

$$\sim_P := \{(x, y) \mid \exists Y \in P \quad (x \in Y \wedge y \in Y)\}$$

then "$\sim_P$" is a equivalence class in "$(A, \sim_P)$".

1	$\quad\vert\ x \in A$
2	$\quad\vert\ x \in \bigcup P$
3	$\quad\vert\ \exists Y (x \in Y \wedge Y \in P)$
4	$\quad\vert\quad\vert\ x \in Y \wedge Y \in P$
5	$\quad\vert\quad\vert\ \exists Y \in P \quad (x \in Y \wedge x \in Y)$
6	$\quad\vert\quad\vert\ x \sim_P x$
7	$\quad\vert\ x \sim_P x$
8	$\forall x \in A \quad x \sim_P x$

1	$\quad\vert\ x, y \in A$
2	$\quad\vert\ x \sim_P y$
3	$\quad\vert\ \exists Y \in P \quad (x \in Y \wedge y \in Y)$
4	$\quad\vert\quad\vert\ Y \in P \wedge x \in Y \wedge y \in Y$
5	$\quad\vert\quad\vert\ \exists Y \in P \quad (y \in Y \wedge x \in Y)$
6	$\quad\vert\quad\vert\ y \sim_P x$
7	$\quad\vert\ y \sim_P x$
8	$\forall x, y \in A \quad x \sim_P y \to y \sim_P x$

1	$x, y, z \in A$
2	$x \sim_P y$
3	$y \sim_P z$
4	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$
5	$\exists Y \in P \quad (y \in Y \wedge z \in Y)$
6	$\quad Y \in P \wedge x \in Y \wedge y \in Y$
7	$\quad\quad Y' \in P \wedge y \in Y' \wedge z \in Y'$
8	$\quad\quad Y \cap Y' \neq \varnothing$
9	$\quad\quad Y = Y'$
10	$\quad\quad Y \in P \wedge x \in Y \wedge z \in Y$
11	$\quad\quad \exists Y \in P \quad (x \in Y \wedge z \in Y)$
12	$\quad\quad x \sim_P z$
13	$\quad x \sim_P z$
14	$x \sim_P z$
15	$\forall x, y, z \in A \quad x \sim_P y \wedge y \sim_p z \to x \sim_P z$

5.7 Orderings

definition 5.7.1. ***(partial) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a relation). we say "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" if it's reflexive,antisymmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \preceq x$$

$$(antisymmetric) : \forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y$$

$$(transitive) : \forall x, y, z \in A \quad x \preceq y \wedge y \preceq z \to x \preceq z$$

definition 5.7.2. ***(partial) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a relation). we say "$\prec$" is a "(partial) Strict order relation" in "$\mathfrak{A}$" if it's irreflexive, and transitive.*

$$(irreflexive) : \forall x \in A \quad x \not\prec x$$

$$(transitive) : \forall x, y, z \in A \quad x \prec y \wedge y \prec z \to x \prec z$$

by two conditions bellow we can proof asymmetric too.

$$(asymmetric) : \forall x, y \in A \quad x \prec y \to y \not\prec x$$

now we will show you relation between "(partial) Order relation" and "(partial) Strict order relation".

Theorem 5.7.1. .

1. *if "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(partial) Strict order relation".

1	$\quad x \in A$
2	$\quad\quad x \prec x$
3	$\quad\quad x \neq x$
4	$\quad\quad \perp$
5	$\quad x \nprec x$
6	$\forall x \in A \quad x \nprec x$

1	$\quad x, y, z \in A$
2	$\quad\quad x \prec y$
3	$\quad\quad y \prec z$
4	$\quad\quad x \preceq y$
5	$\quad\quad y \preceq z$
6	$\quad\quad x \preceq z$
7	$\quad\quad x \neq y$
8	$\quad\quad y \neq z$
9	$\quad\quad\quad x = z$
10	$\quad\quad\quad y \preceq x$
11	$\quad\quad\quad x = y$
12	$\quad\quad\quad \perp$
13	$\quad\quad x \neq z$
14	$\quad\quad x \prec z$
15	$\forall x, y, z \in A \quad x \prec z$

2. *if "$\prec$" is a "(partial) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(partial) order relation".

1	$\quad x \in A$
2	$\quad x = x$
3	$\quad x \preceq x$
4	$\forall x \in A \quad x \preceq x$

$$\begin{array}{ll}
1 & \mid\; x, y \in A \\
2 & \mid\;\mid\; x \preceq y \wedge y \preceq x \\
3 & \mid\;\mid\;\mid\; x \prec y \wedge y \prec x \\
4 & \mid\;\mid\;\mid\; \perp \\
5 & \mid\;\mid\;\mid\; x = y \\
6 & \mid\;\mid\;\mid\; x \prec y \wedge y = x \\
7 & \mid\;\mid\;\mid\; x = y \\
8 & \mid\;\mid\;\mid\; x = y \wedge y \prec x \\
9 & \mid\;\mid\;\mid\; x = y \\
10 & \mid\;\mid\;\mid\; x = y \wedge y = x \\
11 & \mid\;\mid\;\mid\; x = y \\
12 & \mid\;\mid\; x = y \\
13 & \forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y
\end{array}$$

$$\begin{array}{ll}
1 & \mid\; x, y, z \in A \\
2 & \mid\;\mid\; x \preceq y \wedge y \preceq z \\
3 & \mid\;\mid\;\mid\; x \prec y \wedge y \prec z \\
4 & \mid\;\mid\;\mid\; x \prec z \\
5 & \mid\;\mid\;\mid\; x \preceq z \\
6 & \mid\;\mid\;\mid\; z \prec y \wedge y = z \\
7 & \mid\;\mid\;\mid\; x \prec z \\
8 & \mid\;\mid\;\mid\; x \preceq z \\
9 & \mid\;\mid\;\mid\; z = y \wedge y \prec z \\
10 & \mid\;\mid\;\mid\; x \prec z \\
11 & \mid\;\mid\;\mid\; x \preceq z \\
12 & \mid\;\mid\;\mid\; z = y \wedge y = z \\
13 & \mid\;\mid\;\mid\; x = z \\
14 & \mid\;\mid\;\mid\; x \preceq z \\
15 & \mid\;\mid\; x \preceq z \\
16 & \forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y
\end{array}$$

definition 5.7.3. ***(Total-Linear) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is* *a Order relation). we say "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \preceq y \vee y \preceq x$$

definition 5.7.4. ***(Total-Linear) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a Strict order relation). we say "$\prec$" is a "(Total-Linear) Strict order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$$

Theorem 5.7.2. *.*

1. *if "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(Total-Linear) Strict order relation".*
   Proof Is Left As An Exercise To The Reader.😃

2. *if "$\prec$" is a "(Total-Linear) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(Total-Linear) order relation".*
   Proof Is Left As An Exercise To The Reader.😃

Lemma 5.7.1. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B, C \subseteq A$ prove*

1. $\exists b \in B \quad \forall x \in B \quad b \preceq x / \therefore \exists! b \in B \quad \forall x \in B \quad b \preceq x$
   Proof Is Left As An Exercise To The Reader.😃*. in this case we say "B" has least element or Minimum or "$HaveLeast(B)$"*

2. $\exists c \in C \quad \forall x \in C \quad x \preceq c / \therefore \exists! c \in C \quad \forall x \in C \quad x \preceq c$
   Proof Is Left As An Exercise To The Reader.😃*. in this case we say "C" has greatest element or has Maximum or "$HaveGreatest(C)$"*

Theorem 5.7.3. ***Minimum and Maximum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid B \text{ has least element}\}$ and $\mathcal{C} := \{C \subseteq A \mid C \text{ has greatest element}\}$ we define like this*

$$min_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times V \mid \forall x \in B \quad b \preceq x\}$$

$$max_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times V \mid \forall x \in C \quad x \preceq c\}$$

prove these two relations are two functions "$min_{\mathfrak{A}} : \mathcal{B} \to V$" and "$max_{\mathfrak{A}} : \mathcal{C} \to V$".

definition 5.7.5. ***Minimal and Maximal*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we define "Minimal and Maximal" like this*

$$Minimal_{\mathfrak{A}}(b) : \forall x \in A \quad x \not\prec b$$

i.e. "b" is the minimal in "$\mathfrak{A}$".

$$Maximal_{\mathfrak{A}}(c) : \forall x \in A \quad c \not\prec x$$

i.e. "c" is the maximal in "$\mathfrak{A}$".

definition 5.7.6. ***Well ordering*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\mathfrak{A}$ " is "Well ordered" when every nonempty subset of "A " has a "Minimum " i.e.*

$$WellOrdered(\mathfrak{A}) : \forall X \subseteq A \quad X \neq \varnothing \to \exists y \in X \quad \forall x \in X \quad y \preceq x$$

definition 5.7.7. ***Lower and Upper Bound*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). $B, C \subseteq A$. we define "Lower and Upper Bound" like this*

$$Lowerb_{\mathfrak{A}}(b, B) : \forall x \in B \quad b \preceq x$$

i.e. "b" is the lowerbound of B in "$\mathfrak{A}$".

$$Upperb_{\mathfrak{A}}(c, C) : \forall x \in C \quad x \preceq c$$

i.e. "C" is the upperbound of B in "$\mathfrak{A}$".

if the stucture clear to us we don't write stucture index "$min(\_), Minimal(\_), Lowerb(\_, \_)$". and we don't mension it.

Lemma 5.7.2. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B, C \subseteq A$ prove*

1. $\exists b \in A \quad (\forall x \in B \quad b \preceq x \wedge \forall x \in A \quad Lowerb_{\mathfrak{A}}(x, B) \to x \preceq b) / \therefore \exists! b \in A \quad (\forall x \in B \quad b \preceq x \wedge \forall x \in A \quad Lowerb_{\mathfrak{A}}(x, B) \to x \preceq b)$.
   Proof Is Left As An Exercise To The Reader.😃
2. $\exists c \in A \quad (\forall x \in C \quad x \preceq c \wedge \forall x \in A \quad Upperb_{\mathfrak{A}}(x, C) \to c \preceq x) / \therefore \exists! c \in A \quad (\forall x \in C \quad x \preceq c \wedge \forall x \in A \quad Upperb_{\mathfrak{A}}(x, C) \to c \preceq x)$.
   Proof Is Left As An Exercise To The Reader.😃

Theorem 5.7.4. ***Infimum and Supremum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid \exists b \in A \quad \forall x \in B \quad b \preceq x\}$ and $\mathcal{C} := \{C \subseteq A \mid \exists c \in A \quad \forall x \in C \quad x \preceq c\}$ we define like this*

$$inf_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times V \mid \forall x \in B \quad b \preceq x \wedge \forall x \in A \quad Lowerb_{\mathfrak{A}}(x, B) \to x \preceq b\}$$

$$sup_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times V \mid \forall x \in C \quad x \preceq c \wedge \forall x \in A \quad Upperb_{\mathfrak{A}}(x, C) \to c \preceq x\}$$

prove these two relations are two functions "$inf_{\mathfrak{A}} : \mathcal{B} \to V$" and "$sup_{\mathfrak{A}} : \mathcal{C} \to V$".

Theorem 5.7.5. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B \subseteq A$ prove*

1. $min_{\mathfrak{A}}(B) = b / \therefore inf_{\mathfrak{A}}(B) = b$.
   Proof Is Left As An Exercise ToThe Reader.😃
2. $inf_{\mathfrak{A}}(B) = b, b \in B / \therefore min_{\mathfrak{A}}(B) = b$.
   Proof Is Left As An Exercise ToThe Reader.😃

this theorem holds for "max, sup" too.

This page intentionally left blank.

Chapter 6

Natural numbers

6.1 Introduction to Natural numbers

definition 6.1.1. ***Natural numbers*** *we define "$0, 1, 2, 3, \ldots$" like this.*

$$
\begin{aligned}
&0 := \varnothing \\
&1 := \{0\} = \varnothing \cup \{\varnothing\} = \{\varnothing\} \\
&2 := \{0, 1\} = \{\varnothing\} \cup \big\{\{\varnothing\}\big\} = \big\{\varnothing, \{\varnothing\}\big\} \\
&3 := \{0, 1, 2\} = \Big\{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\Big\} \\
&\vdots
\end{aligned}
$$

definition 6.1.2. ***Successor function*** *is a function "$s : V \to V$" defined like this.*

$$s(x) := x \cup \{x\}$$

Axiom 6.1.1. ***The Axiom of Infinity*** *we define a set "Inductive" like this.*

$$Ind(X) : \varnothing \in X \land \forall y(y \in X \to s(y) \in X)$$

notice that "$Ind(X)$" is not a proposition. so we can't use weak comprehension. this axiom says it is a proposition. so "Ind" is a set and it's a nonepmty set because we can prove "$V \in Ind$". so we can define "$\bigcap Ind$".

definition 6.1.3. ***Set of Natural numbers($\mathbb{N}$)***

$$\mathbb{N} := \bigcap Ind$$

Lemma 6.1.1. *prove.*

1. $\mathbb{N} = \{x \mid \underbrace{\forall Y\Big(\big(\underbrace{\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)}_{Ind(Y)}\big) \to x \in Y\Big)}_{\phi(x)}\}$

2. $Ind(\mathbb{N})$
   Proof Is Left As An Exercise To The Reader.😃

notice that now "$\phi(x)$" is a proposition. so "$\{x \mid \forall Y\Big(\big(\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)\big) \to x \in Y\Big)\}$" is a set.

6.2 Properties of Natural numbers

Theorem 6.2.1. ***Induction*** *$P(0), \forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big) / \therefore \forall n \in \mathbb{N} \quad P(n)$*
proof.

1	$P(0)$
2	$\forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big)$
3	$\quad\mid\; n \in \mathbb{N}$
4	$\quad\mid\; \forall n\Big(P(n) \to P(s(n))\Big)$
5	$\quad\mid\; Ind(P)$
6	$\quad\mid\; n \in \bigcap Ind$
7	$\quad\mid\; \forall X(Ind(X) \to n \in X)$
8	$\quad\mid\; Ind(P) \to P(n)$
9	$\quad\mid\; P(n)$
10	$n \in \mathbb{N} \to P(n)$
11	$\forall n \in \mathbb{N} \quad P(n)$

definition 6.2.1. ***Order in Natural numbers($\mathbb{N}$)***

$$<:= \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x \in y\}$$

Theorem 6.2.2. ***Order of Natural numbers($\mathbb{N}$)*** *prove "$(\mathbb{N}, <)$*

1. *is lineary ordered.*
2. *is Well ordered.*
3. $\Big(\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x < y\Big)$ *(doesn't have right endpoint)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 6.2.3. ***Induction ,second vrsion*** $\forall k < n\Big(P(k) \to P(n)\Big) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

6.3 The Recurtion theorem

definition 6.3.1. ***Tuple (new definition)-Finite sequence*** *is a function "$a : n \to V$" s.t. $n \in \mathbb{N}$ we know we can show this by "$(a_i)_{i\in n}$" we can also show it by "$(a_0, a_1, a_2, ..., a_{n-1})$"*

definition 6.3.2. ***Sequence*** *is a function "$a : \mathbb{N} \to V$" we know we can show this by "$(a_i)_{i\in\mathbb{N}}$" we can also show it by "$(a_i)_{i=0}^{\infty}$"*

now we can redefine "Cartesian product". we define finite Cartesian product "$\prod_{i\in n} A_i := \{(a_i)_{i\in n} \mid a_i \in A_i\}$" and define infinite Cartesian product "$\prod_{i\in\mathbb{N}} A_i := \{(a_i)_{i\in\mathbb{N}} \mid a_i \in A_i\}$" and define Cartesian product in general way "$\prod_{i\in I} A_i := \{(a_i)_{i\in I} \mid a_i \in A_i\}$". and you can easily see that from the definition of "B^A", "A^n" is the set of all n-length finite sequences of $a \in A$ and "$A^{\mathbb{N}}$" is the set of all infinite sequences of $a \in A$.

Theorem 6.3.1. ***Recurtion*** *for any set "A" and "$a \in A$" and any function "$g : A \times \mathbb{N} \to A$" there is a unique infinite sequence "$f : \mathbb{N} \to A$" s.t.*

1. $f(0) = a$.
2. $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$.

Claim 1 : Existing of "f"

1 $\quad F := \{t \mid \forall k, m \in \mathbb{N} \quad 0 < k < m \to \Big(\big(t : s(m) \to A\big) \wedge t(0) = a \wedge t\big(s(k)\big) = g\big(t(k), k\big)\Big)\}$

2 $\quad f := \bigcup F$

Claim 2 : "f" is a function

1 $\quad Func(f)$

Claim 3 : "$Dom(f) = \mathbb{N}$" and "$Range(f) \subseteq A$"

Claim 4 : "f" satisfies conditions (1) and (2)

Theorem 6.3.2. ***(Parametric version) recurtion*** *for any set "A, P" and "$a : P \to A$" and any function "$g : P \times A \times \mathbb{N} \to A$" there is a unique function "$f : P \times \mathbb{N} \to A$" s.t.*

1. $f(p, 0) = a(p)$
2. $\forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(n), n)$

Claim 1 : Existing of "f"
Claim 2 : "f" is a function
Claim 3 : "$Dom(f) = \mathbb{N}$" and "$Range(f) \subseteq A$"

Claim 4 : "f" satisfies conditions (1) and (2)

6.4 Arithmetic of Natural numbers

we want to introduce some functions “$\bigcirc : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$” to you. we use “$m \bigcirc n$” instead of “$\bigcirc(m, n)$”.

Theorem 6.4.1. ***Summation*** *there is a unique function “$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$” s.t.*

1. $\forall m \in \mathbb{N} \quad m + 0 = m$
2. $\forall m, n \in \mathbb{N} \quad m + s(n) = s(m + n)$

Proof Is Left As An Exercise To The Reader.😃

Lemma 6.4.1. *prove for “$(\mathbb{N}, +)$”*

1. $\forall m, n, k \in \mathbb{N} \quad (m + n) + k = m + (n + k)$ *(assosiative $+$)*
2. $\forall m \in \mathbb{N} \quad m + 0 = 0 + m = m$ *(identity $+$)*
3. $\forall m, n \in \mathbb{N} \quad m + n = n + m$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 6.4.2. ***Multiplication*** *there is a unique function “$\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$” s.t.*

1. $\forall m \in \mathbb{N} \quad m \times 0 = 0$
2. $\forall m, n \in \mathbb{N} \quad m \times s(n) = m \times n + m$

Proof Is Left As An Exercise To The Reader.😃

we denote also “Multiplication” with “$m \cdot n$” and “mn” too.

Lemma 6.4.2. *prove for “$(\mathbb{N}, +, \times)$”*

1. $\forall m, n, k \in \mathbb{N} \quad (mn)k = m(nk)$ *(assosiative $\times$)*
2. $\forall m \in \mathbb{N} \quad m \cdot 1 = 1 \cdot m = m$ *(identity $\times$)*
3. $\forall m, n \in \mathbb{N} \quad mn = 0 \to m = 0 \vee n = 0$ *(zero divisor free $\times$)*
4. $\forall m, n \in \mathbb{N} \quad mn = nm$ *(commutative(abelian) $\times$)*
5. $\forall m, n, k \in \mathbb{N} \quad \Big(m(n + k) = mn + mk\Big) \wedge \Big((m + n)k = mk + nk\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 6.4.3. *prove for “$(\mathbb{N}, +, \times, <)$”*

1. $\forall m, n, k \in \mathbb{N} \quad m < n \to m + k < n + k$
2. $\forall m, n, k \in \mathbb{N} \quad m < n \wedge 0 < k \to mk < nk$

Proof Is Left As An Exercise To The Reader.😃

Theorem 6.4.3. ***Exponents*** *there is a unique function “$(\_)^{(\_)} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$” s.t.*

1. $\forall m \in \mathbb{N} \quad m^0 = 1$
2. $\forall m, n \in \mathbb{N} \quad m^{s(n)} = m^n \cdot m$

Proof Is Left As An Exercise To The Reader.😃

6.5 Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitary functions and constants and doesn't have meaning we defined before). with these properties.

1. $\forall m, n \in N \quad m + s(n) = s(m + n)$
2. $\forall m \in N \quad m + 0 = m$
3. $\forall m \in N \quad m + 0 = m$
4. $\forall m, n \in N \quad m + s(n) = s(m + n)$
5. $\forall m \in N \quad m \cdot 0 = 0$
6. $\forall m, n \in N \quad m \cdot s(n) = m \cdot n + m$
7. $\phi(0) \wedge \forall n \in N \quad \Big(\phi(n) \to \phi(s(n))\Big) \to \forall n \in N \quad \phi(n)$ (this is true for arbitary "ϕ")

we call "$\mathfrak{N}$" a model of "Peano arithmetic".

Lemma 6.5.1. *prove for "$(N, +)$"*

1. $\forall m, n, k \in \mathbb{N} \quad (m + n) + k = m + (n + k)$ *(assosiative +)*
2. $\forall m \in \mathbb{N} \quad m + 0 = 0 + m = m$ *(identity +)*
3. $\forall m, n \in \mathbb{N} \quad m + n = n + m$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😀

Lemma 6.5.2. *prove for "$(N, +, \times)$"*

1. $\forall m, n, k \in \mathbb{N} \quad (mn)k = m(nk)$ *(assosiative ×)*
2. $\forall m \in \mathbb{N} \quad m \cdot 1 = 1 \cdot m = m$ *(identity ×)*
3. $\forall m, n \in \mathbb{N} \quad mn = 0 \to m = 0 \vee n = 0$ *(zero divisor free ×)*
4. $\forall m, n \in \mathbb{N} \quad mn = nm$ *(commutative(abelian) ×)*
5. $\forall m, n, k \in \mathbb{N} \quad \Big(m(n + k) = mn + mk\Big) \wedge \Big((m + n)k = mk + nk\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😀

as you can see "$(\mathbb{N}, +, \times, s, 0)$" is a model of peano arithmetic. so properties are true for them. you can add "Exponents" axioms and prove more.

Lemma 6.5.3. *prove for "$(N, +, \times, (\_)^{(\_)}, s, 0)$"*

1. $\forall m, n, k \in \mathbb{N} \quad m^n \cdot m^k = m^{n+k}$
2. $\forall m, n, k \in \mathbb{N} \quad (m^n)^k = m^{nk}$

Proof Is Left As An Exercise To The Reader.😀

This page intentionally left blank.

Chapter 7

From Natural numbers to Complex numbers

7.1 Integers“$\mathbb{Z}$”

we learned about “$\mathbb{N}$” before and we constructed structure “$(\mathbb{N}, +, \cdot, <)$”. if you assume “$\mathbb{N} \times \mathbb{N}$” we can define an equivalence relation “$(n_1, m_1) \sim (n_2, m_2) : n_1 + m_2 = n_2 + m_1$”(check it's an equivalence relation).

definition 7.1.1. ***Integers“$\mathbb{Z}$”*** *is a set*

$$\mathbb{Z} := \mathbb{N} \times \mathbb{N} / \sim$$

we denote it's members with “$z_0, x, \ldots$”

in this system “$0 := [(0, 0)]$” and “$\forall n \in \mathbb{N} \quad n := [(n, 0)]$” and “$\forall n \in \mathbb{N} \quad -n := [(0, n)]$”. in general “$\forall m, n \in \mathbb{N} \quad m - n := [(m, n)]$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z}$”.

Theorem 7.1.1. ***Summation*** *there is a unique function “$+_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$” s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] +_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 + m_2, n_1 + n_2)]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.1.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$” s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] \times_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 m_2 + n_1 n_2, m_1 n_2 + m_2 n_1)]$$

Proof Is Left As An Exercise To The Reader.😃

definition 7.1.2. ***Order in*** $\mathbb{Z}$

$$[(m_1, n_1)] <_{\mathbb{Z}} [(m_2, n_2)] : m_1 + n_2 < m_2 + n_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}}$", "$+$" instead of "$+_{\mathbb{Z}}$" and "$\times$" instead of "$\times_{\mathbb{Z}}$".

Lemma 7.1.1. *prove.*

1. $/ \therefore \forall m, n \in \mathbb{N} \quad m <_{\mathbb{Z}} n \leftrightarrow m < n$
2. $/ \therefore \forall m, n \in \mathbb{N} \quad m +_{\mathbb{Z}} n = m + n$
3. $/ \therefore \forall m, n \in \mathbb{N} \quad m \times_{\mathbb{Z}} n = m \times n$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 7.1.2. *prove for "$(\mathbb{Z}, +)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall z \in \mathbb{Z} \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Z} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 7.1.3. *prove for "$(\mathbb{Z}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall z \in \mathbb{Z} \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{Z} \quad xy = 0 \to x = 0 \lor y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in \mathbb{Z} \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Z} \quad \Big(x(y + z) = xy + xz\Big) \land \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.1.3. ***Order of Integers($\mathbb{Z}$)*** *prove for "$(\mathbb{Z}, +, \times, <_{\mathbb{Z}})$ "*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad x < y\Big) \land \Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z} \quad x < y \to x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z} \quad x < y \land 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

7.2 Rational numbers "$\mathbb{Q}$"

we learned about "$\mathbb{Z}$" before and we constructed structure "$(\mathbb{Z}, +, \cdot, <)$". if you assume "$\mathbb{Z} \times (\mathbb{Z} - \{0\})$" we can define an equivalence relation "$(y_1, x) \sim (y_2, y) : y_1 \cdot y = y_2 \cdot x$"(check it's an equivalence relation).

definition 7.2.1. ***Rational numbers*"$\mathbb{Q}$"** *is a set*

$$\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} - \{0\}) / \sim$$

we denote it's members with "$q_0, q_1, \ldots$"

in this system "$0 := [(0,1)]$" and "$1 := [(1,1)]$" and "$\forall x \in \mathbb{Z} \quad x := [(x,1)]$" and "$\forall x \in \mathbb{Z} \quad \frac{1}{x} := [(1,x)]$". in general "$\forall x, y \in \mathbb{Z} \quad \frac{x}{y} := [(x,y)]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$".

Theorem 7.2.1. ***Summation*** *there is a unique function "$+_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] +_{\mathbb{Q}} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.2.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] \times_{\mathbb{Q}} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

definition 7.2.2. ***Order in*** $\mathbb{Q}$.
if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Q}}$", "$+$" instead of "$+_{\mathbb{Q}}$" and "$\times$" instead of "$\times_{\mathbb{Q}}$".

Lemma 7.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z} \quad z <_{\mathbb{Q}} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z} \quad z +_{\mathbb{Q}} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z} \quad z \times_{\mathbb{Q}} y = z \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 7.2.2. *prove for "$(\mathbb{Q}, +)$"*

1. $\forall x, y, z \in \mathbb{Q} \quad (x + y) + z = x + (y + z)$ *(assosiative $+$)*
2. $\forall x \in \mathbb{Q} \quad x + 0 = 0 + x = x$ *(identity $+$)*
3. $\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x + y = y + x = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{Q} \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 7.2.3. *prove for "$(\mathbb{Q}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{Q} \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall x \in \mathbb{Q} \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{Q} - \{0\} \quad \exists y \in \mathbb{Q} \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{Q} \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Q} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.2.3. ***Order of Rational numbers($\mathbb{Q}$)*** *prove for "$(\mathbb{Q}, <_{\mathbb{Q}})$"*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q} \quad \exists z \in \mathbb{Q} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q} \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

7.3 Real numbers "$\mathbb{R}$"(Dedekind Method)

we learned about "$\mathbb{Q}$" before and we constructed structure "$(\mathbb{Q}, +, \cdot, <)$".

definition 7.3.1. ***Cut*** *is a set*

$$Cut := \{(X, Y) \mid Part(\{X, Y\}, \mathbb{Q}) \wedge (\forall y \in X \quad \forall x \quad x < y \to x \in X)\}$$

notice that because $\mathbb{Q}$ is dense it's not possible both "X" has greatest element and "Y" has least element.

definition 7.3.2. ***Dedekind cut***

$$Dedekind(X, Y) : Cut(X, Y) \wedge \forall x \in X \quad \exists y \in X \quad x < y$$

Dedekind cuts like "(X, Y)" are two types.

1. those "Y" has a least element we denote them by "$[min(Y)] := (X, Y)$"
2. those "Y" has not a least element we call them "Gaps"

definition 7.3.3. ***Real numbers "$\mathbb{R}$"*** *is a set*

$$\mathbb{R} := \{(X, Y) \mid Dedekind(X, Y)\}$$

we denote it's members with "$r_0, r_1, \ldots$"

in this system "$0 := [0]$" and "$1 := [1]$" and "$\forall q \in \mathbb{Q} \quad q := [q]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$".

Theorem 7.3.1. ***Summation*** *there is a unique function “$+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$” s.t.*

$$(X,Y) +_{\mathbb{R}} (Z,U) = (\{x+y \mid x \in X \wedge y \in Y\}, \{x+y \mid x \in Z \wedge y \in U\})$$

Proof Is Left As An Exercise To The Reader.😀

definition 7.3.4. ***Order in*** $\mathbb{R}$ *.*

$$(X,Y) <_{\mathbb{R}} (Z,U) : X \subseteq Z$$

Theorem 7.3.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$” s.t.* ($\mathbb{Q}^{\leq 0} := \{x \in \mathbb{Q} \mid x \leq 0\} \quad , \quad \mathbb{Q}^{0\leq} := \{x \in \mathbb{Q} \mid 0 \leq x\}$)
if $0 <_{\mathbb{R}} (X,Y), (Z,U)$

$$(X,Y) \times_{\mathbb{R}} (Z,U) = (\mathbb{Q}^{\leq 0} \cup \{xy \mid x \in X \wedge y \in Z \wedge x, y > 0\}, \{xy \mid x \in Z \wedge y \in U\})$$

if $0 <_{\mathbb{R}} (X,Y), 0 >_{\mathbb{R}} (Z,U)$

$$(X,Y) \times_{\mathbb{R}} (Z,U) = (\{xy \mid x \in Y \wedge y \in Z\}, \{xy \mid x \in X \wedge y \in U \wedge x > 0 \wedge y < 0\} \cup \mathbb{Q}^{0\leq})$$

if $0 >_{\mathbb{R}} (X,Y), 0 <_{\mathbb{R}} (Z,U)$

$$(X,Y) \times_{\mathbb{R}} (Z,U) = (\{xy \mid x \in X \wedge y \in U\}, \{xy \mid x \in Y \wedge y \in Z \wedge x < 0, y > 0\} \cup \mathbb{Q}^{0\leq})$$

if $0 >_{\mathbb{R}} (X,Y), (Z,U)$

$$(X,Y) \times_{\mathbb{R}} (Z,U) = (\mathbb{Q}^{\leq 0} \cup \{xy \mid x \in Y \wedge y \in U \wedge x, y < 0\}, \{xy \mid x \in X \wedge y \in Z\})$$

Proof Is Left As An Exercise To The Reader.😀

this lemma let us use “$<$” instead of “$<_{\mathbb{R}}$”, “$+$” instead of “$+_{\mathbb{R}}$” and “$\times$” instead of “$\times_{\mathbb{R}}$”.

Lemma 7.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader.😀

we denote “Multiplication” with “$x \cdot y$” and “xy” too.

Lemma 7.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x+y)+z = x+(y+z)$ *(assosiative +)*
2. $\forall x \in \mathbb{R} \quad x+0 = 0+x = x$ *(identity +)*
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x+y = y+x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R} \quad x+y = y+x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 7.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ *(assosiative* $\times$*)*
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ *(identity* $\times$*)*
3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ *(inverse* $\times$*)*
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ *(commutative(abelian)* $\times$*)*
5. $\forall x, y, z \in \mathbb{R} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive* $\times$*)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.3.3. ***Linear order of Real numbers(***$\mathbb{R}$***)*** *prove "*$(\mathbb{R}, <_{\mathbb{R}})$

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R} \quad Uppreb(X, y)) \to (\exists y \in \mathbb{R} \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R} \quad x < y \to x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

7.4 Real numbers "$\mathbb{R}$"(Cauchy Method)

we want to make real numbers in another way. we use the "$\mathbb{R}$" name for this method too in the end you will see these two structures are isomprpphic.

before we we get to lesson define function "$|\_| : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not fimiliar to this notation it means "$|\_| := \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid (0 \leq x \to y = x) \wedge (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}$" before and we constructed structure "$(\mathbb{Q}, +, \cdot, <)$". now assume $\mathbb{Q}^{\mathbb{N}}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}$). we don't want it all we just want a subset.

definition 7.4.1. ***Cauchy sequence*** *is a sequence like "*$(a_i)_{i \in \mathbb{N}} \in \mathbb{Q}^{\mathbb{N}}$*" s.t. elements of it get infinitly close to each other i.e.*

$$Cauchy(a_i)_{i \in \mathbb{N}} : \forall \epsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \epsilon$$

we usually give “N” natural numbers but it’s not necessary.

we define an equivalence relation “$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}} : \forall\epsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \epsilon$”(check it’s an equivalence relation).

definition 7.4.2. ***Real numbers*** *“$\mathbb{R}$” is a set*

$$\mathbb{R} := Cauchy/\sim$$

we denote it’s members with “$r_0, r_1, \ldots$”

in this system “$0 := [(0)_{i\in\mathbb{N}}]$” and “$1 := [(1)_{i\in\mathbb{N}}]$” and “$\forall q \in \mathbb{Q} \quad q := [(q)_{i\in\mathbb{N}}]$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$”.

Theorem 7.4.1. ***Summation*** *there is a unique function “$+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$” s.t.*

$$[(a_i)_{i\in\mathbb{N}}] +_{\mathbb{R}} [(b_i)_{i\in\mathbb{N}}] = [(a_i + b_i)_{i\in\mathbb{N}}]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.4.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$” s.t.*

$$[(a_i)_{i\in\mathbb{N}}] \times_{\mathbb{R}} [(b_i)_{i\in\mathbb{N}}] = [(a_i b_i)_{i\in\mathbb{N}}]$$

Proof Is Left As An Exercise To The Reader.😃

definition 7.4.3. ***Order in*** $\mathbb{R}$ *.*

$$[(a_i)_{i\in\mathbb{N}}] <_{\mathbb{R}} [(b_i)_{i\in\mathbb{N}}] : \exists\epsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \epsilon$$

this lemma let us use “$<$” instead of “$<_{\mathbb{R}}$”, “$+$” instead of “$+_{\mathbb{R}}$” and “$\times$” instead of “$\times_{\mathbb{R}}$”.

Lemma 7.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote “Multiplication” with “$x \cdot y$” and “xy” too.

Lemma 7.4.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall x \in \mathbb{R} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 7.4.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ *(assosiative $\times$)*
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{R} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.4.3. ***Linear order of Real numbers($\mathbb{R}$)*** *prove "$(\mathbb{R}, <_{\mathbb{R}})$*

1. *is lineary ordered.*
2. $\Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \rightarrow x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R} \quad Uppreb(X, y)) \rightarrow (\exists y \in \mathbb{R} \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R} \quad x < y \rightarrow x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader.😃

7.5 Complex numbers"$\mathbb{C}$"

we learned about "$\mathbb{R}$" before and we constructed structure "$(\mathbb{R}, +, \cdot, <)$".

definition 7.5.1. ***Complex numbers"$\mathbb{C}$"*** *is a set*

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}$$

we denote it's members with "$z_0, x, \ldots$"

in this system "$0 := (0,0)$" and "$\forall n \in \mathbb{R} \quad r := (r, 0)$" . in general "$\forall x, y \in \mathbb{R} \quad x + iy := (x, y)$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$".

Theorem 7.5.1. ***Summation*** *there is a unique function "$+_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$" s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) +_{\mathbb{C}} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 7.5.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$" s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) \times_{\mathbb{C}} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.😃

this lemma let us use "$+$" instead of "$+_{\mathbb{C}}$" and "$\times$" instead of "$\times_{\mathbb{C}}$".

Lemma 7.5.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R} \quad x +_{\mathbb{C}} y = x + y$
2. $/ \therefore \forall x, y \in \mathbb{R} \quad x \times_{\mathbb{C}} y = x \times y$

Proof Is Left As An Exercise To The Reader.😀

we denote “Multiplication” with “$z \cdot y$” and “zy” too.

Lemma 7.5.2. *prove for “$(\mathbb{C}, +)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (x + y) + z = x + (y + z)$ *(assosiative +)*
2. $\forall z \in \mathbb{C} \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{C} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😀

Lemma 7.5.3. *prove for “$(\mathbb{C}, +, \times)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (xy)z = x(yz)$ *(assosiative ×)*
2. $\forall z \in \mathbb{C} \quad z \cdot 1 = 1 \cdot z = z$ *(identity ×)*
3. $\forall x, y \in \mathbb{C} \quad xy = yx$ *(commutative(abelian) ×)*
4. $\forall x, y, z \in \mathbb{C} \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😀

This page intentionally left blank.

Chapter 8

Finite and Infinite sets

8.1 Equipotent and Order in sets

we talked about "Isomorphism" before now assume two Structures $\mathfrak{A} := (A)$ and $\mathfrak{B} := (B)$. because these Structures have only a set "Isomorphism" between them only means there is a bijection function $f : A \xrightarrow[surj]{inj} B$.

definition 8.1.1. ***Equipotent*** *in situation above we say "A" and "B" are "Equipotent". we denote it with "$A \cong B$" i.e.*

$$A \cong B : \exists f (f : A \xrightarrow[surj]{inj} B)$$

Lemma 8.1.1. *"$\cong$" is a equivalence relation in "$(V, \cong)$"*
Proof Is Left As An Exercise To The Reader.😃

so "$V/\cong$" is a partision for "V" and every partition equals to a $[A]_{\cong}$ we denote it by $[A]$.

definition 8.1.2. ***Equipotent*** *if there is a function $f : A \xrightarrow{inj} B$. in situation above we say "A" is less than or equipotent "B". we denote it with "$A \precsim B$" i.e.*

$$A \precsim B : \exists f (f : A \xrightarrow{inj} B)$$

similarly we define "$[A] \preceq [B] : \forall X \in [A], Y \in [B] \quad X \precsim Y$"

Lemma 8.1.2. $A \precsim B \therefore / \therefore [A] \preceq [B]$
Proof Is Left As An Exercise To The Reader.😃

Theorem 8.1.1. ***Schroeder-Bernstein*** $A \precsim B, B \precsim A / \therefore A \cong B$
Proof Is Left As An Exercise To The Reader.😃

Lemma 8.1.3. *"$\preceq$" is a order relation in "$(V/\cong, \preceq)$"*
Proof Is Left As An Exercise To The Reader.😃

8.2 Finite sets

definition 8.2.1. *if there is a "$n \in \mathbb{N}$" s.t. $A \cong n$ we call it **"Finite"**($Finite(A)$). if a set is not finite we call it **"Infinite"**($Infinite(A)$).*

Lemma 8.2.1. $/ \therefore \forall n \in \mathbb{N} \quad X \subsetneq n \to X \ncong n$
Proof Is Left As An Exercise To The Reader.😃

Corollary 8.2.1. .

1. $\forall m, n \in \mathbb{N} \quad m \neq n \to m \ncong n$
2. $\forall m, n \in \mathbb{N} \quad A \cong m \wedge A \cong n \to m = n$
3. $Infinite(\mathbb{N})$
4. $\forall m, n \in \mathbb{N} \quad m \prec n \leftrightarrow m < n$

Proof Is Left As An Exercise To The Reader.😃

corollary (4) let us use "$<$" instead of "$\prec$"

Lemma 8.2.2. $Finite(X), Y \subseteq X / \therefore Finite(Y)$
Proof Is Left As An Exercise To The Reader.😃

8.3 Countable sets

definition 8.3.1. *set "A" is **"Countable"** if "$A \cong \mathbb{N}$". set "A" is **"At most countable"** if "$A \precsim \mathbb{N}$"*

Lemma 8.3.1. $Countable(X), Infinite(Y), Y \subseteq X / \therefore Countable(Y)$
Proof Is Left As An Exercise To The Reader.😃

Corollary 8.3.1. *a set is at most countable iff it's finite or countable.*

Lemma 8.3.2. *range of infinite set is atmost countable.*
Proof Is Left As An Exercise To The Reader.😃

Lemma 8.3.3. *union of two countable set is countable.*
Proof Is Left As An Exercise To The Reader.😃

Corollary 8.3.2. *finite union of countable sets is countable.*

Lemma 8.3.4. *if "A" and "B" are countable sets then "$A \times B$" is countable.*
Proof Is Left As An Exercise To The Reader.😃

Corollary 8.3.3. *finite cartesian product of countable sets is countable. and "A^n" is countable for any "A" countable and "$n > 0$".*

Lemma 8.3.5. *if "$(A_i)_{i \in \mathbb{N}}$" is family of countable sets with a family of sequences "$(a_i)_{i \in \mathbb{N}}$" s.t. "$\forall n \in \mathbb{N} \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} A_n$" then "$\bigcup_{i \in \mathbb{N}} A_i$" is countable.*
Proof Is Left As An Exercise To The Reader.😃

Lemma 8.3.6. *if "A" is countable set then "$\bigcup_{n\in\mathbb{N}} A^n$" is countable.*
Proof Is Left As An Exercise To The Reader. 😃

This page intentionally left blank.

Chapter 9

Ordinals

This page intentionally left blank.

Chapter 10

Cardinals

This page intentionally left blank.

Chapter 11

Axiom of chioce

This page intentionally left blank.

Chapter 12

hyperreal numbers($*\mathbb{N}, *\mathbb{Z}, *\mathbb{Q}, *\mathbb{R}$) and surreal numbers

This page intentionally left blank.

Chapter 13

fantacy numbers($\mathbb{N}(Ord), \mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord)$)

This page intentionally left blank.

Chapter 14

New beginning for calculus

This page intentionally left blank.

Chapter 15

Lebesgue Intergral

This page intentionally left blank.

Chapter 16

New beginning for Probability theory